YEAR 10 MATHEMATICS · STAGE 5 PATH

Year 10
Circle Geometry

Lesson and Homework Pack

Revision	Angles, isosceles triangles, congruence, similarity and proof	Year 8 · 3F, 3K, 6B, 6C, 8E, 8G, 8H
Lesson C1	Chords, radii and angles	Past papers: 10 marks · 15 min + timed practice
Lesson C2	Angles at the centre and circumference	Past papers: 17 marks · 25 min
Lesson C3	Tangents and secants	Past papers: 18 marks · 25 min
Review	Circle geometry review	Past papers: 14 marks · 20 min

This pack covers the circle geometry of the NSW syllabus (MA5-CIR-P-01): the terminology, the chord and angle properties, tangents and secants, and their proofs. Each lesson runs in the same order: **concept** → **tutor demo** → **your attempt** → **worked example** to keep for homework. The homework comes in three bands, **Easy**, **Medium** and **Hard**, and each lesson ends with a **past-paper set** to do under exam conditions. Answers and full past-paper solutions are in the separate Homework Answers booklet.

The lessons follow the AMSI TIMES module *Circle Geometry* (Years 9–10, Module 26). The homework also uses the Maths Pathway worksheets and Year 10 yearly examinations from NSW schools.

REVISION

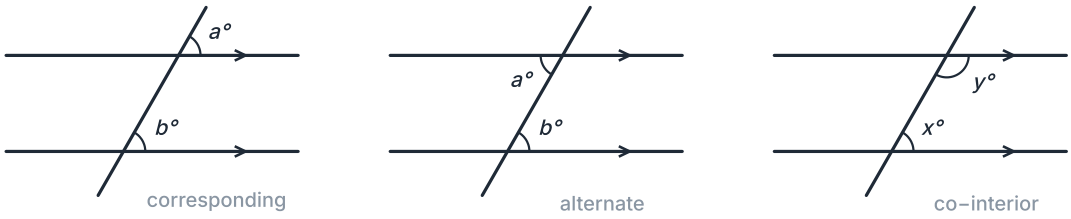
Angles, triangles, congruence and proof

Circle geometry is built on the geometry of Years 7 to 9: angle facts with parallel lines, the angle sum and exterior angle of a triangle, isosceles triangles, the congruence and similarity tests, and proofs written as a chain of steps, each with a reason. These pages review that work from the Year 8 textbook (Cambridge Maths NSW Stage 4 Year 8, chapters 3, 6 and 8). Every circle theorem in this pack is proved with these tools.

SUMMARY · keep these pages open while you work

TRANSVERSALS AND PARALLEL LINES (YEAR 8, 6B)

A **transversal** is a line cutting two or more other lines. When a transversal crosses two or more lines, pairs of angles can be **corresponding** (in matching positions), **alternate** (on opposite sides of the transversal and inside the other two lines), **co-interior** (on the same side of the transversal and inside the other two lines), vertically opposite, or angles on a straight line. Lines are parallel if they do not intersect. Parallel lines are marked with the same number of arrows.

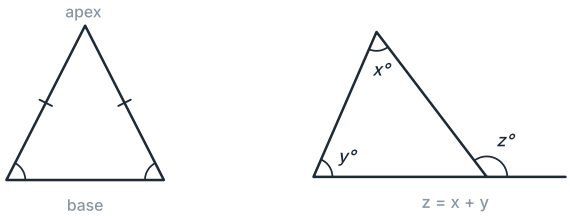


When two parallel lines are cut by a transversal:

- the corresponding angles are equal ($a = b$);
- the alternate angles are equal ($a = b$);
- the co-interior angles are supplementary, that is, they sum to $180°$ ($x + y = 180$).

TRIANGLES (YEAR 8, 6C)

- A triangle has 3 sides, 3 vertices and 3 interior angles. Sides with the same number of dashes are of equal length.
- Triangles are classified by side lengths (scalene, isosceles, equilateral) and by interior angles (acute-angled, right-angled, obtuse-angled).
- The angle sum of a triangle is $180°$.
- **The exterior angle theorem:** the exterior angle of a triangle is equal to the sum of the two opposite interior angles ($z = x + y$).

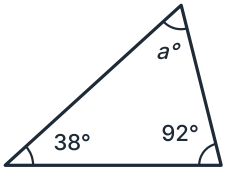


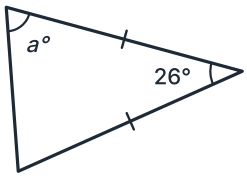
ISOSCELES TRIANGLES

In an isosceles triangle the two equal sides meet at the **apex**. The angles opposite the equal sides, the **base angles**, are equal. In circle geometry, two radii of the same circle are always equal, so a triangle with two radii as sides is isosceles. This is used in almost every proof in this pack.

WORKED EXAMPLE **Year 8 Example 3 (6C)**

Find the value of a in these triangles.





SOLUTION

a $a + 38 + 92 = 180$ (angle sum of triangle)

$$a + 130 = 180$$

$$\therefore a = 50$$

b $a + a + 26 = 180$

$$2a + 26 = 180 \quad \text{(angle sum of a triangle)}$$

$$2a = 154$$

$$\therefore a = 77$$

STEPS THE BOOK SKIPS: YEAR 8 EXAMPLE 3 In **a**, $38 + 92 = 130$ and $180 - 130 = 50$. In **b**, the angles opposite the equal sides are equal, so both base angles are $a°$.

CONGRUENT TRIANGLES (YEAR 8, 8E)

There are four 'tests' that can be used to decide if two triangles are congruent. Two triangles are congruent when:

- the **three sides of a triangle** are respectively equal to the **three sides of another triangle** (**SSS test**);
- **two sides and the included angle** of a triangle are respectively equal to **two sides and the included angle of another triangle** (**SAS test**);
- **two angles and one side** of a triangle are respectively equal to **two angles and the matching side of another triangle** (**AAS test**);
- the **hypotenuse and a second side of a right-angled triangle** are respectively equal to the **hypotenuse and a second side of another right-angled triangle** (**RHS test**).

Corresponding (matching) parts of two figures have the same geometric properties. A congruence statement can be written using the symbol $\equiv$. For example,

$\triangle ABC \equiv \triangle FDE$ is read as '$\triangle ABC$ is congruent to $\triangle FDE$'. In a congruence statement, vertices are named in matching order.

SIMILAR TRIANGLES (YEAR 8, 8G)

Two triangles are similar when:

- two angles of a triangle are equal to two angles of another triangle;
- three sides of a triangle are proportional to three sides of another triangle;
- two sides of a triangle are proportional to two sides of another triangle, and the included angles are equal;
- the hypotenuse and a second side of a right-angled triangle are proportional to the hypotenuse and second side of another right-angled triangle.

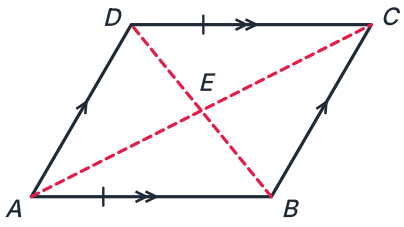
If two triangles $\triangle ABC$ and $\triangle DEF$ are similar, we write $\triangle ABC \,|||\, \triangle DEF$. The common ratio of matching sides is the **scale factor**.

PROPERTIES OF THE SPECIAL QUADRILATERALS (YEAR 8, 8H)

- **Parallelogram:** a quadrilateral with both pairs of opposite sides parallel. The opposite sides are equal, the opposite angles are equal, and the diagonals bisect each other.
- **Rhombus:** a parallelogram with two adjacent sides equal in length. All sides are equal, and the diagonals bisect each other at right angles and bisect the vertex angles.
- **Rectangle:** a parallelogram with a right angle. All angles at the vertices are $90°$, and the diagonals are equal and bisect each other.
- **Square:** a rectangle with two adjacent sides equal. The diagonals are equal, bisect each other at right angles and bisect the vertex angles.
- **Kite:** two pairs of adjacent sides equal. One diagonal bisects the other, and the diagonals are perpendicular.

WORKED EXAMPLE **Year 8 Example 13 (8H)**

Prove that the diagonals of a parallelogram bisect each other and give reasons.



SOLUTION

$\angle BAE = \angle DCE$ (alternate angles in parallel lines)

$\angle ABE = \angle CDE$ (alternate angles in parallel lines)

$AB = CD$ (given equal side lengths)

$\therefore \triangle ABE \equiv \triangle CDE$ (AAS)

$\therefore BE = DE$ and $AE = CE$

$\therefore$ Diagonals bisect each other.

THE SHAPE OF A GEOMETRIC PROOF

Every line of a proof is a statement followed by its reason in brackets. Name the two triangles first, list three matching facts, state the test, and then use ‘matching sides (or angles) of congruent triangles’ to reach the result. The two triangles above are congruent by AAS, and since $\triangle ABE$ and $\triangle CDE$ are congruent, the corresponding sides are equal. The same pattern proves the chord theorems of Lesson C1.

PYTHAGORAS’ THEOREM AND SECTORS (YEAR 8, 3K AND 3F)

Pythagoras’ theorem. The hypotenuse is the longest side of a right-angled triangle, opposite the right angle. The square of the hypotenuse is the sum of the squares of the lengths of the other two shorter sides: $a^2 + b^2 = c^2$.

Sectors. A sector is formed by dividing a circle with two radii. A sector’s area is determined by calculating a fraction of the area of a circle with the same radius. The fraction is $\dfrac{\theta}{360}$, so

$$\text{Sector area} = \frac{\theta}{360} \times \pi r^2.$$

FROM TOPIC 11: THE AREA OF A TRIANGLE

The area of a triangle with two sides a and b and included angle C is $\text{Area} = \frac{1}{2}ab\sin C$. One past-paper question in Lesson C1 uses it.

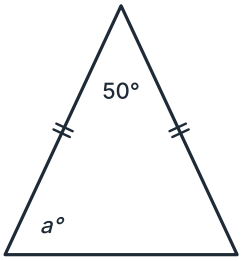
WARM-UP · answers are in the answers booklet

Each question is from the Year 8 textbook exercise named, with some parts left out.

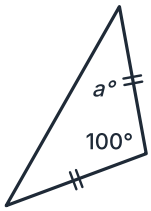
1 Year 8 Exercise 6C Q6

Find the value of a in these isosceles triangles.

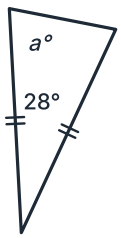
a



b



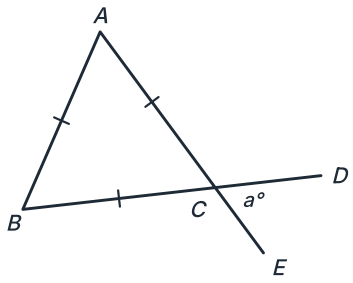
c



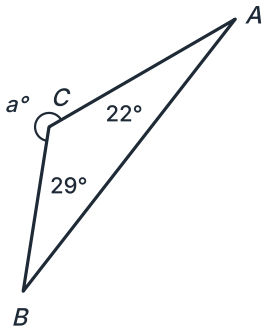
2 Year 8 Exercise 6C Q10

Find the value of a in these diagrams, giving reasons.

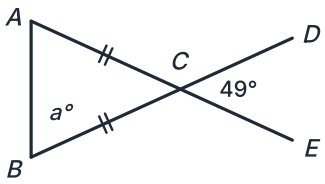
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b



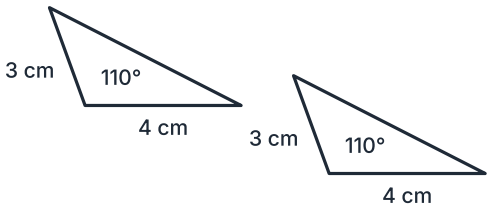
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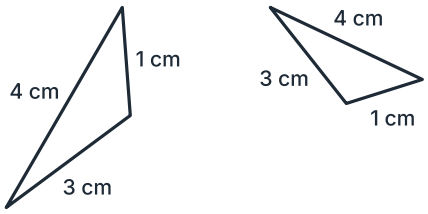
3 Year 8 Exercise 8E Q3

Which of the tests (SSS, SAS, AAS or RHS) would you choose to test the congruence of these triangles?

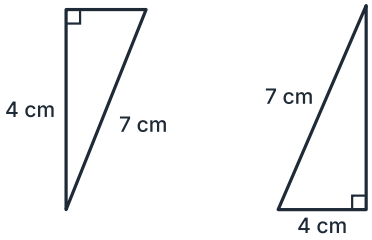
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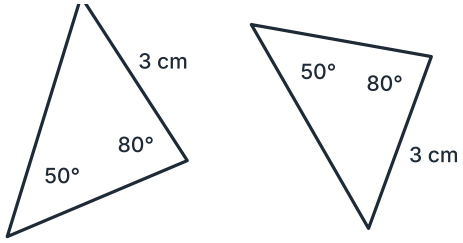
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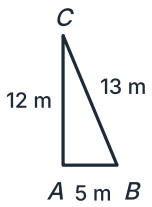
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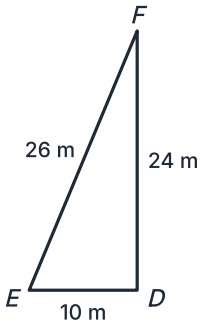


4 Year 8 Exercise 8G Q3

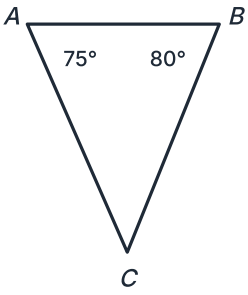
Decide, with reasons, why these pairs of triangles are similar.

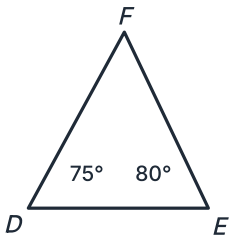
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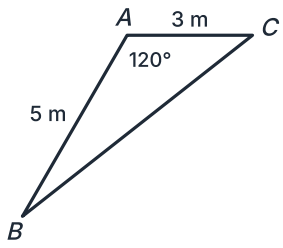


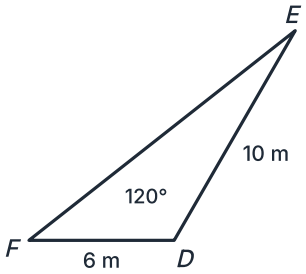
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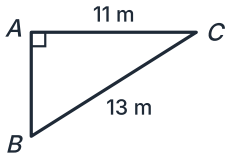


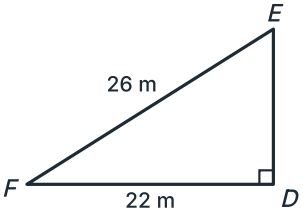
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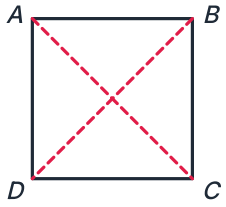
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5 Year 8 Exercise 8H Q5

Given $ABCD$ is a square, complete the following proof to show that the diagonals of a square are equal.



In $\triangle ABC$ and $\triangle BCD$:

$\angle ABC =$ _____ (angles in a _____ are equal)

$AB = BC$ (_____)

_____ is common.

$\therefore \triangle ABC \equiv \triangle BCD$ (S __ __)

$\therefore AC =$ _____ (matching sides in _____ triangles are equal)

and the _____ of a square are equal in length.

LESSON C1

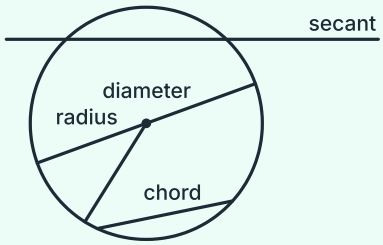
Chords, radii and angles

This lesson sets up the language of circles: radius, chord, diameter, secant, arc, sector and segment, and what it means for an angle to be subtended. Then come the first theorems, each proved with isosceles and congruent triangles: the line from the centre to the midpoint of a chord is perpendicular to it, equal chords subtend equal angles at the centre, and an angle in a semicircle is a right angle. The teaching follows the AMSI TIMES module *Circle Geometry*, pages 5 to 11.

CONCEPT 1 · circles and their parts

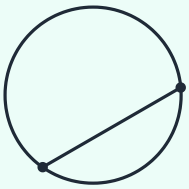
RADII AND CHORDS (AMSI, PAGE 5)

We begin by recapitulating the definition of a circle and the terminology used for circles. Throughout this module, all geometry is assumed to be within a fixed plane.

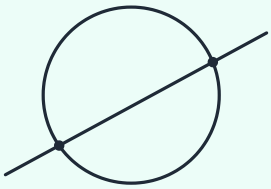


- A **circle** is the set of all points in the plane that are a fixed distance (**the radius**) from a fixed point (**the centre**).
- Any **interval** joining a point on the circle to the centre is called a **radius**. By the definition of a circle, any two radii have the same length. Notice that the word ‘radius’ is being used to refer both to these intervals and to the common length of these intervals.
- An interval joining two points on the circle is called a **chord**.
- A chord that passes through the centre is called a **diameter**. Since a diameter consists of two radii joined at their endpoints, every diameter has length equal to twice the radius. The word ‘diameter’ is used to refer both to these intervals and to their common length.
- A line that cuts a circle at two distinct points is called a **secant**. Thus a chord is the interval that the circle cuts off a secant, and a diameter is the interval cut off by a secant passing through the centre of a circle.

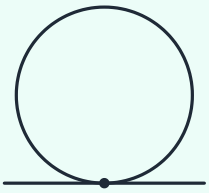
CHORDS, SECANTS AND TANGENTS (MATHS PATHWAY)



A **chord** is a line segment between two points on a circle



A **secant** is a line which meets the circle at *two* points, and extends beyond the circle



A **tangent** is a line which intersects the circle at *one* point

Note: A chord which is extended into a full line is a secant.

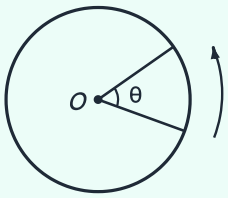
TANGENTS COME LATER

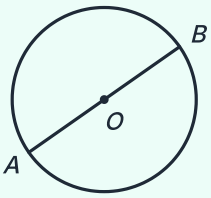
Lesson C3 studies tangents in detail. For now you only need to recognise one: a tangent touches the circle at exactly one point, while a secant cuts it at two.

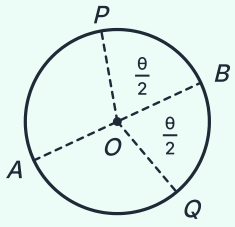
SYMMETRIES OF A CIRCLE (AMSI, PAGE 6)

Circles have an abundance of symmetries:

- A circle has every possible rotation symmetry about its centre, in that every rotation of the circle about its centre rotates the circle onto itself.
- If AOB is a diameter of a circle with centre O, then the reflection in the line AOB reflects the circle onto itself. Thus every diameter of the circle is an axis of symmetry.





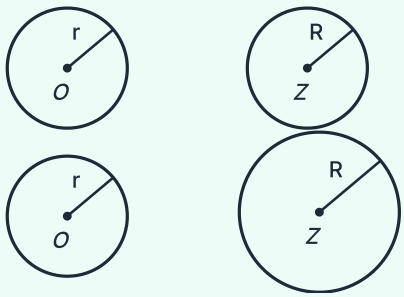


As a result of these symmetries, any point P on a circle can be moved to any other point Q on the circle. This can be done by a rotation through the angle $\theta = \angle POQ$ about the centre. It can also be done by a reflection in the diameter AOB bisecting $\angle POQ$. Thus every point on a circle is essentially the same as every other point on the circle – no other figure in the plane has this property except for lines.

CONGRUENCE AND SIMILARITY OF CIRCLES (AMSI, PAGE 6)

Any two circles with the same radius are congruent – if one circle is moved so that its centre coincides with the centre of the other circle, then it follows from the definition that the two circles will coincide.

More generally, any two circles are similar – move one circle so that its centre coincides with the centre of the other circle, then apply an appropriate enlargement so that it coincides exactly with the second circle.



A circle forms a curve with a definite length, called the **circumference**, and it encloses a definite area. The similarity of any two circles is the basis of the definition of π, the ratio of the circumference and the diameter of any circle. We saw in the module, *The Circles* that if a circle has radius r, then

$$\text{circumference of the circle} = 2\pi r \qquad \text{and} \qquad \text{area of the circle} = \pi r^2$$

SEGMENTS, AND THE WORD 'SUBTEND' (AMSI, PAGE 8)

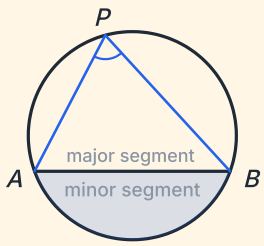
A chord AB of a circle divides the circle into two **segments**. If AB is a diameter, the two congruent segments are called **semicircles** – the word 'semicircle' is thus used both for the semicircular arc, and for the segment enclosed by the arc and the diameter. Otherwise, the two segments are called a **major segment** and a **minor segment**.

The word 'subtend' literally means 'holds under', and is often used in geometry to describe an angle.

Suppose that we have an interval or arc AB and a point P not on AB. Join the intervals AP and BP to form the angle $\angle APB$. We say that the angle $\angle APB$ is the angle subtended by the interval or arc AB at the point P.

OUR OWN PICTURE: SEGMENTS AND A SUBTENDED ANGLE

The chord AB cuts the circle into the shaded minor segment and the unshaded major segment. The angle $\angle APB$, marked in blue, is the angle subtended by the chord AB at the point P. Think of the chord AB as 'held under' the two arms PA and PB.



TUTOR DEMO **Naming lines and measuring circles**

1 Specify whether each of the following is a chord, secant, tangent, or none.

a  **b**  **c**  **d**  **e**  **f** 

2 A circle has radius 6 cm. Find its diameter, and its circumference and area in exact form.

YOUR TURN **Same ideas**

1 Specify whether each of the following is a chord, secant, tangent, or none.

a  **b**  **c**  **d**  **e**  **f** 

2 A circle has radius 4.5 cm. Find its diameter, and its circumference and area in exact form.

WORKED EXAMPLE **Maths Pathway, Chords, Secants and Tangents, Q1(a)–(f): keep for homework**

Specify whether each of the following is a chord, secant, tangent, or none.

a  **b**  **c**  **d**  **e**  **f** 

SOLUTION

a Secant **b** Chord **c** None **d** Tangent **e** None **f** Chord

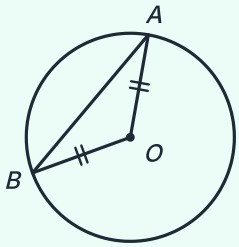
STEPS THE SOURCE SKIPS: Q1(A)–(F)

In **a** the line goes on beyond both points where it cuts the circle, so it is a secant. In **c** the line misses the circle. In **e** the interval joins a point on the circle to the centre: it is a radius, not a chord. In **f** the interval joins two points of the circle through the centre: it is a diameter, which is a chord.

CONCEPT 2 · radii and chords, the perpendicular from the centre

RADII AND CHORDS (AMSI, PAGE 7)

Let AB be a chord of a circle not passing through its centre O. The chord and the two equal radii OA and BO form an isosceles triangle whose base is the chord. The angle $\angle AOB$ is called the **angle at the centre subtended by the chord**.

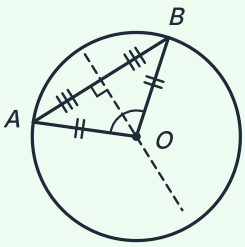


In the module, *Rhombuses, Kites and Trapezia* we discussed the axis of symmetry of an isosceles triangle. Translating that result into the language of circles:

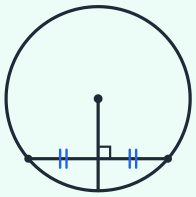
THEOREM

Let AB be a chord of a circle with centre O. The following three lines coincide:

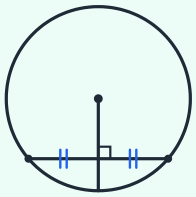
- The bisector of the angle $\angle AOB$ at the centre subtended by the chord.
- The interval joining O and the midpoint of the chord AB.
- The perpendicular bisector of the chord AB.



TWO THEOREMS RELATED TO CHORDS AND RADII (MATHS PATHWAY)



If a radius is perpendicular to a chord, then this radius also bisects the chord.



If a radius bisects a chord, then this radius is also perpendicular to the chord.

STEPS THE SOURCE SKIPS: PROVING THE THREE LINES ARE THE SAME LINE

The AMSI module quotes this theorem from an earlier module, so here are the three short proofs. In each, M is a point of AB, and $\triangle OAM$ and $\triangle OBM$ share the side OM, with $OA = OB$ (radii).

1. **Midpoint gives perpendicular.** If $AM = BM$, then $\triangle OAM \equiv \triangle OBM$ (SSS). The worked example below finishes this proof.
2. **Perpendicular gives midpoint.** If $OM \perp AB$, then $\angle OMA = \angle OMB = 90°$, so $\triangle OAM \equiv \triangle OBM$ (RHS). So $AM = BM$ (matching sides of congruent triangles).
3. **Angle bisector gives both.** If $\angle AOM = \angle BOM$, then $\triangle OAM \equiv \triangle OBM$ (SAS). So $AM = BM$, and $\angle OMA = \angle OMB$, which are adjacent angles on the straight line AB, so each is $90°$.

So each of the three lines passes through O and the midpoint of AB. The perpendicular bisector of AB passes through O because $OA = OB$: any point that is the same distance from A and from B lies on the perpendicular bisector of AB. Only one line passes through two given points, so the three lines coincide.

TRIGONOMETRY AND CHORDS (AMSI, PAGE 7)

Constructions with radii and chords give plenty of opportunity for using trigonometry.

This exercise shows that sine can be regarded as the length of the semichord AM in a circle of radius 1, and cosine as the perpendicular distance of the chord from the centre. Until modern times, tables of sines were compiled as tables of chords or semichords, and the name ‘sine’ is conjectured to have come in a complicated and confused way from the Indian word for semichord.

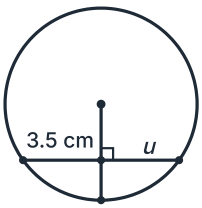
WHICH EXERCISE?

‘This exercise’ is AMSI Exercise 2, Medium homework Question 9 below. In a circle of radius r, Pythagoras’ theorem finds a chord, the radius or the distance of a chord from the centre.

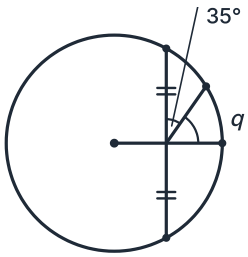
TUTOR DEMO **The perpendicular from the centre**

Find the value of the missing variable, giving reasons.

a



b

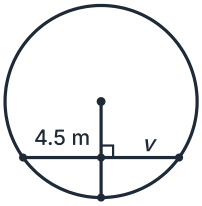


c A chord of length 24 cm is drawn in a circle of radius 13 cm. Find the perpendicular distance of the chord from the centre.

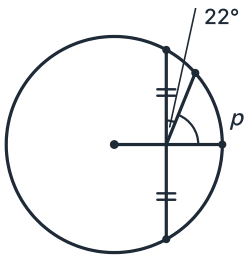
YOUR TURN **Same method**

Find the value of the missing variable, giving reasons.

a



b



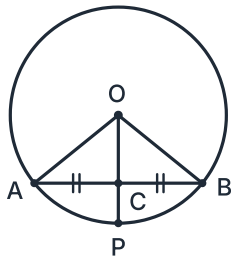
c A chord of length 16 cm is drawn in a circle of radius 10 cm. Find the perpendicular distance of the chord from the centre.

WORKED EXAMPLE **Maths Pathway, Chords, Secants and Tangents, Q4: keep for homework**

Use congruent triangles to prove that OP is perpendicular to AB.

a Prove that $\Delta AOC \equiv \Delta BOC$.

b Use what you proved in part (a) to prove that OP is perpendicular to AB.



SOLUTION

a Proof: In ΔAOC and ΔCOB, we compare corresponding angles and sides.

$AO = BO$ (Radii of the same circle)

$AC = BC$ (Given)

OC is a common side in both ΔAOC and ΔBOC.

Therefore $\Delta AOC \equiv \Delta BOC$ by SSS.

b

$$\begin{aligned} \angle ACO + \angle BCO &= 180^\circ && (\angle ACO \text{ and } \angle BCO \text{ form a straight angle}) \\ \angle ACO &= \angle BCO && (\text{Corresponding angles in congruent triangles}) \\ \angle ACO + \angle ACO &= 180^\circ && (\text{Substitution}) \\ \angle ACO &= \frac{180^\circ}{2} \\ &= 90^\circ \end{aligned}$$

Therefore OP is perpendicular to AB.

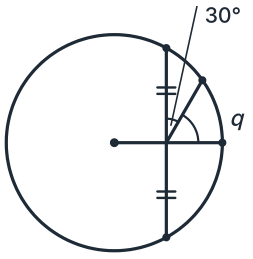
Note: Your working may look different, but the reasoning must be the same.

STEPS THE SOURCE SKIPS: Q4

In **b**, $2 \times \angle ACO = 180^\circ$, so $\angle ACO = 90^\circ$. Since C lies on OP, the line OP meets AB at right angles.

WORKED EXAMPLE **Maths Pathway, Chords, Secants and Tangents, Q5(b): keep for homework**

Find the value of the missing variable, giving reasons.



SOLUTION

The radius bisects the chord, so it is also perpendicular to the chord.

Therefore

$$\begin{aligned} q + 30^\circ &= 90^\circ \\ q &= 90^\circ - 30^\circ \\ &= 60^\circ \end{aligned}$$

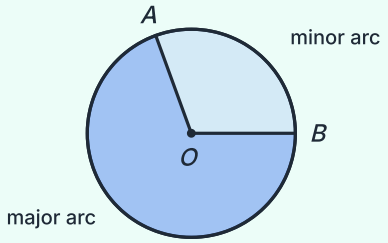
STEPS THE SOURCE SKIPS: Q5(B)

The two marks on the chord say that the radius cuts it into two equal parts. So the angle between the chord and the radius is $90°$. The $30°$ angle and q together make up this right angle.

CONCEPT 3 · arcs, sectors and equal chords

ARCS AND SECTORS (AMSI, PAGES 7 AND 8)

Let A and B be two different points on a circle with centre O. These two points divide the circle into two **opposite arcs**. If the chord AB is a diameter, then the two arcs are called **semicircles**. Otherwise, one arc is longer than the other – the longer arc is called the **major arc** AB and the shorter arc is called the **minor arc** AB.



Now join the radii OA and OB. The reflex angle $\angle AOB$ is called the **angle subtended by the major arc** AB. The non-reflex angle $\angle AOB$ is called the **angle subtended by the minor arc** AB, and is also the angle subtended by the chord AB.

The two radii divide the circle into two sectors, called correspondingly the **major sector** OAB and the **minor sector** OAB.

It is no surprise that equal chords and equal arcs both subtend equal angles at the centre of a fixed circle. The result for chords can be proven using congruent triangles, but congruent triangles cannot be used for arcs because they are not straight lines, so we need to identify the transformation involved.

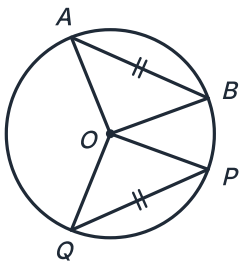
THEOREM

a Equal chords of a circle subtend equal angles at the centre.
b Equal arcs of a circle subtend equal angles at the centre.

WORKED PROOF **From the text (AMSI, page 8): keep for homework**

PROOF

a In the diagram to the right, $\triangle AOB \equiv \triangle POQ$ (SSS)
so $\angle AOB = \angle POQ$ (matching angles of congruent triangles)



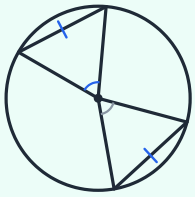
b Rotate the circle so that the arc PQ coincides with the arc AB or BA. Then the angles $\angle POQ$ and $\angle AOB$ coincide, and hence are equal.

STEPS THE SOURCE SKIPS: THE SSS TEST IN PART A

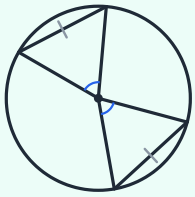
The three pairs of equal sides are $OA = OP$ (radii), $OB = OQ$ (radii) and $AB = PQ$ (the chords are given equal). Every radius of a circle has the same length, which is why two triangles with their apex at the centre are so often congruent.

EQUAL ANGLES AND EQUAL CHORDS (MATHS PATHWAY)

Two theorems related to chords:



In a circle, if two chords are equal, then the angles they subtend at the centre are equal.



In a circle, if two angles subtended at the centre are equal, then the chords which subtend them are equal.

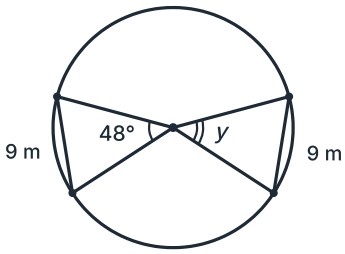
THE SECOND THEOREM IS THE CONVERSE

The second theorem swaps the 'if' part and the 'then' part of the first. You prove it in Medium homework Question 11.

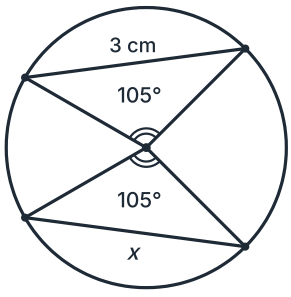
TUTOR DEMO **Equal chords and equal angles**

Find the value of the missing variable, giving reasons.

a



b

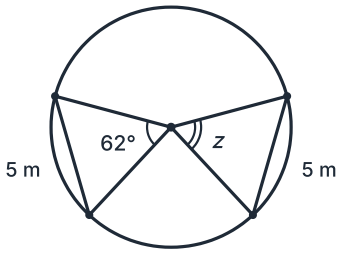


c In part **a**, find the size of each of the other two angles of the triangle that contains the $48°$ angle.

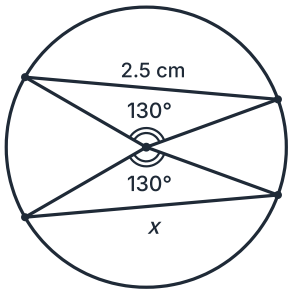
YOUR TURN **Same method**

Find the value of the missing variable, giving reasons.

a



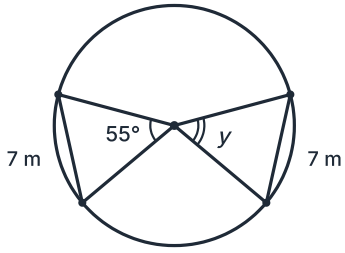
b



c In part **a**, find the size of each of the other two angles of the triangle that contains the $62°$ angle.

WORKED EXAMPLE **Maths Pathway, Chords, Secants and Tangents, Q3(b): keep for homework**

Find the value of the missing variable, giving reasons.



SOLUTION

The chords are equal, so the angles they subtend at the centre are also equal.

Therefore

$y = 55°$

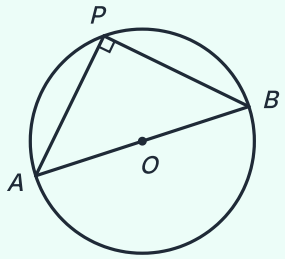
STEPS THE SOURCE SKIPS: Q3(B)

Both chords are 7 m, and each one, with two radii, makes a triangle with its apex at the centre. The $55°$ angle and y are the two angles at the centre, so they are equal.

CONCEPT 4 · angles in a semicircle

ANGLES IN A SEMICIRCLE (AMSI, PAGE 9)

Let AOB be a diameter of a circle with centre O, and let P be any other point on the circle. The angle $\angle APB$ subtended at P by the diameter AB is called an **angle in a semicircle**. This angle is always a right angle – a fact that surprises most people when they see the result for the first time.

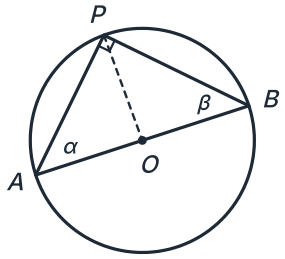


THEOREM

An angle in a semicircle is a right angle.

WORKED PROOF **From the text (AMSI, page 9): keep for homework**

PROOF



Let AB be a diameter of a circle with centre O, and let P be any other point on the circle.

Join the radius PO, and let $\alpha = \angle A$ and $\beta = \angle B$.

The triangles AOP and BOP are isosceles because all radii are equal, so $\angle APO = \alpha$ and $\angle BPO = \beta$ (base angles of isosceles triangles AOP and BOP).

Hence $\alpha + \beta + (\alpha + \beta) = 180°$ (angle sum of $\triangle APB$)

$$2\alpha + 2\beta = 180°$$

$$\alpha + \beta = 90°,$$

so $\angle APB = 90°$, as required.

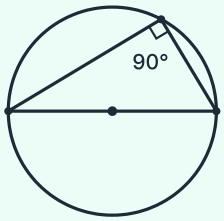
This famous theorem is traditionally ascribed to the Greek mathematician Thales, the first known Greek mathematician.

STEPS THE SOURCE SKIPS: THE PROOF OF THALES' THEOREM

The radius PO splits $\angle APB$ into two parts, so $\angle APB = \angle APO + \angle BPO = \alpha + \beta$. The three angles of $\triangle APB$ are therefore α at A, β at B and $\alpha + \beta$ at P, which is where $\alpha + \beta + (\alpha + \beta)$ comes from.

ANGLES SUBTENDED BY THE DIAMETER (MATHS PATHWAY)

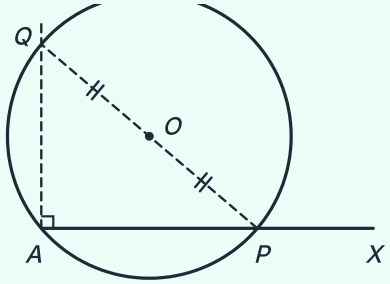
Every angle subtended at a point on the circle by the diameter is $90°$.



CONSTRUCTING A RIGHT ANGLE AT THE ENDPOINT OF AN INTERVAL (AMSI, PAGE 10)

Thales' theorem gives a quick way to construct a right angle at the endpoint of an interval AX.

1. Choose a point O above the interval AX.
2. Draw a circle with centre O through A crossing AX again at P.
3. Draw the radius PO and produce it to a diameter POQ.
4. Join AQ, then $\angle PAQ = 90°$.



WHY THE CONSTRUCTION WORKS

PQ is a diameter and A is on the circle, so $\angle PAQ$ is an angle in a semicircle.

THE CONVERSE THEOREM (AMSI, PAGE 10)

The angle in a semicircle theorem has a straightforward converse that is best expressed as a property of a right-angled triangle:

THEOREM

The circle whose diameter is the hypotenuse of a right-angled triangle passes through all three vertices of the triangle.

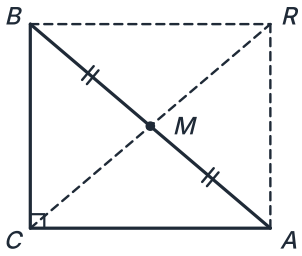
WORKED PROOF **From the text (AMSI, page 10): keep for homework**

PROOF

Let $\triangle ABC$ be right-angled at C, and let M be the midpoint of the hypotenuse AB.

We need to prove that $MC = MA = MB$.

Complete the rectangle $ACBR$. Because $ACBR$ is a rectangle, its diagonals bisect each other and are equal. Hence M is the midpoint of the other diagonal CR, and $AM = BM = CM = RM$.



This is an excellent example of the way ideas in geometry fit together – a significant theorem about circles has been proven using a property of rectangles.

STEPS THE SOURCE SKIPS: THE CONVERSE

Since $MA = MB = MC$, the circle with centre M and radius MA passes through A, B and C. Its diameter is AB, the hypotenuse. A useful consequence: the midpoint of the hypotenuse is the same distance from all three vertices, so $CM = \frac{1}{2}AB$.

TWO PRACTICAL SITUATIONS TO ILLUSTRATE THE CONVERSE THEOREM (AMSI, PAGE 10)

A set of points in the plane is often called a **locus**. The term is used particularly when the set of points is the curve traced out by a moving point. For example, a circle can be defined as the locus of a point that moves so that its distance from some fixed point is constant. The two examples below use the converse of the angle in a semicircle theorem to describe a locus.

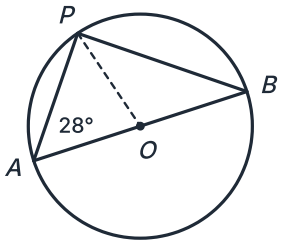
WHERE ARE THE TWO EXAMPLES?

They are AMSI Exercises 4 and 5, homework Questions 12 (Medium) and 15 (Hard) below.

TUTOR DEMO **Angles in a semicircle**

a AB is a diameter of the circle with centre O, and P is a point on the circle with $\angle A = 28°$. Find $\angle APO$, $\angle APB$, $\angle BPO$ and $\angle B$, giving reasons.

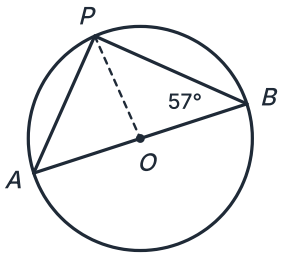
b $\triangle ABC$ is right-angled at C, with $AC = 6$ cm and $BC = 8$ cm. M is the midpoint of AB. Find the length of CM.



YOUR TURN **Same method**

a AB is a diameter of the circle with centre O, and P is a point on the circle with $\angle B = 57°$. Find $\angle BPO$, $\angle APB$, $\angle APO$ and $\angle A$, giving reasons.

b $\triangle ABC$ is right-angled at C, with $AC = 5$ cm and $BC = 12$ cm. M is the midpoint of AB. Find the length of CM.



HOMEWORK · Maths Pathway sheets and AMSI exercises

Each question is from the source named above it. Answers are in the answers booklet.

EASY

1 Maths Pathway, Chords, Secants and Tangents, Q1(g)–(l)

Specify whether each of the following is a chord, secant, tangent, or none.

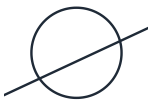
g 

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i 

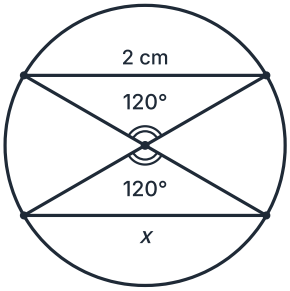
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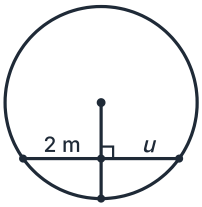
2 Maths Pathway, Chords, Secants and Tangents, Q3(a)

Find the value of the missing variable, giving reasons.



3 Maths Pathway, Chords, Secants and Tangents, Q5(a)

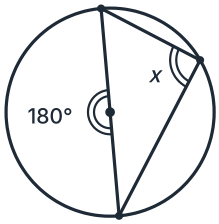
Find the value of the missing variable, giving reasons.



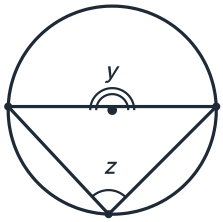
4 Maths Pathway, Arcs and Angles in Circles, Q7

Calculate the values of the variables in the following circle figures.

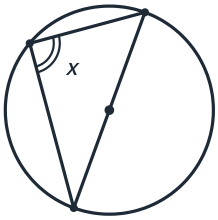
a



b

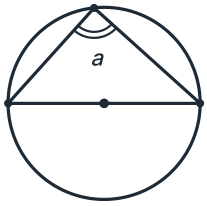


c



5 Maths Pathway, Arcs and Angles in Circles, Q8(a)

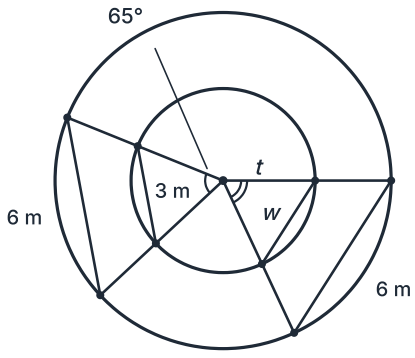
Calculate the values of the variables in the following circle figures, and explain your answer.



MEDIUM

6 Maths Pathway, Chords, Secants and Tangents, Q3(d)

Find the value of the missing variable, giving reasons.



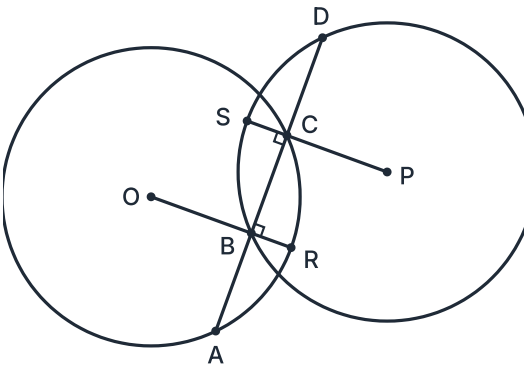
7 Maths Pathway, Chords, Secants and Tangents, Q6(a)

Copy the figure. Then, use the information provided to find the required value. Show your working.

Given information:

- $AB = 4\ \text{mm}$

Find the length of AD.



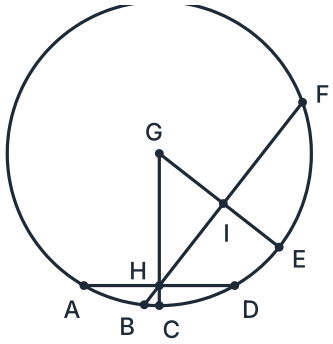
8 Maths Pathway, Chords, Secants and Tangents, Q6(b)

Copy the figure. Then, use the information provided to find the required value. Show your working.

Given information:

- $\angle IHD = 52°$
- $AH = HD$
- $BI = IF$

Find the value of $\angle IGH$.

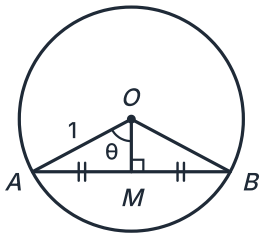


9 AMSI TIMES Circle Geometry, Exercise 2

In the diagram to the right, the interval AM subtends an angle θ at the centre O of a circle of radius 1.

Show that:

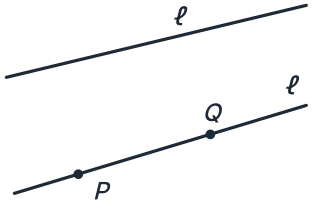
a $AM = \sin\theta$ b $OM = \cos\theta$



10 AMSI TIMES Circle Geometry, Exercise 1

a Identify all translations, rotations and reflections of the plane that map a line ℓ onto itself.

b Which of the transformations in part **a** map a particular point P on ℓ to another particular point Q on ℓ.

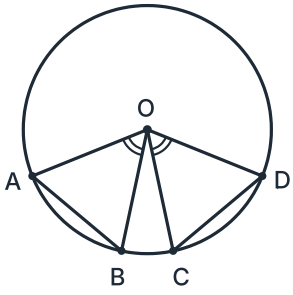


11 Maths Pathway, Chords, Secants and Tangents, Q2

Use congruent triangles to prove that $AB = CD$.

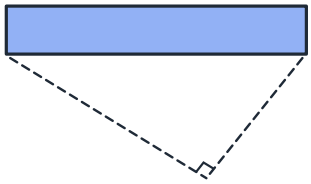
a Prove that $\Delta AOB \equiv \Delta COD$.

b Use what you proved in part (a) to prove that $AB = CD$.



12 AMSI TIMES Circle Geometry, Exercise 4

A photographer is photographing the ornamental front of a building. He wants the two ends of the front to subtend a right angle at his camera. Describe the set of all positions where he can stand.



HARD

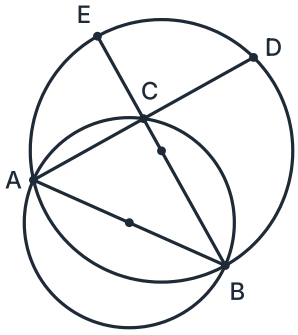
13 Maths Pathway, Chords, Secants and Tangents, Q6(c)

Copy the figure. Then, use the information provided to find the required value. Show your working.

Given information:

- AB is the diameter of the smaller circle.
- BE is the diameter of the larger circle.
- $AB = 5\ \text{m}$
- $BC = 4\ \text{m}$

Find the length of CD.

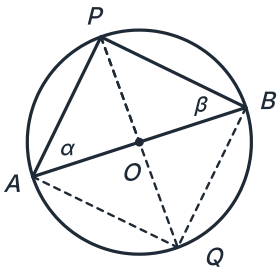


14 AMSI TIMES Circle Geometry, Exercise 3

Thales' theorem can also be proven using the following test for a rectangle developed in the module, *Parallelograms and Rectangles*.

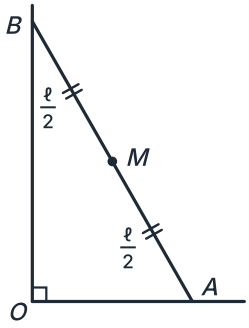
A quadrilateral whose diagonals are equal and bisect each other is a rectangle.

Join the radius PO and produce it to a diameter POQ, then join up the quadrilateral $APBQ$. Explain why $\angle APB$ is a right angle.



15 AMSI TIMES Circle Geometry, Exercise 5

A plank of length ℓ metres is initially resting flush against a wall, but it slips outwards, with its top sliding down the wall and its foot moving at right angles to the wall. What path does the midpoint of the plank trace out?



PAST-PAPER HOMEWORK · mini exam

Suggested time: 15 minutes **10 marks** Calculator allowed · show working · hints are below each question if you get stuck

These four items use only this lesson's ideas. For extra timed practice, not part of the 10 marks, do Questions 13, 14 and 15 above in 15 minutes.

PP 1 **Sydney Boys High School, 2020, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics Stage 5.3, Q40(b) I** **1 mark**

Darth Parker is constructing a Maff Star by intersecting a sphere of radius 100 km with a sphere of radius 40 km such that their centres are 120 km apart.
The smaller sphere is then discarded, leaving a curved dent on the surface of the larger sphere, as shown in Diagram 1 below.

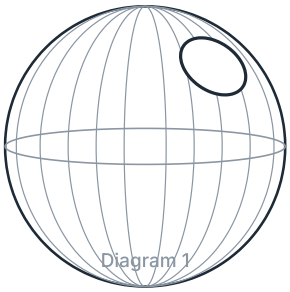


Diagram 1

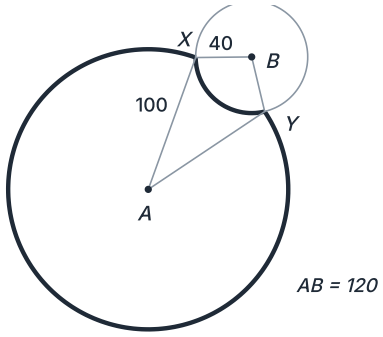


Diagram 2

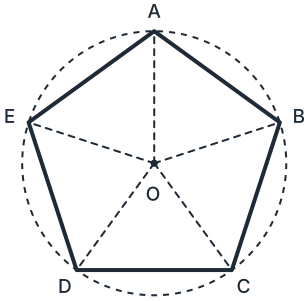
Diagram 2 above shows a cross section of the intersecting spheres, where A and B are the centres of the larger and smaller sphere respectively and X and Y mark where the spheres intersect on the cross section.

I. Given that AB bisects XY, show that $AB \perp XY$.

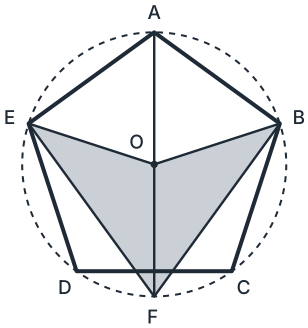
Hint: what do you know about the line AB and the points X and Y?

PP 2 **Sydney Boys High School, 2022, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part A Q9** 2 marks

Ikan Deesine is creating a logo.
Ikan starts by drawing a regular pentagon $ABCDE$.
The vertices of the pentagon lie on the circumference of a circle with centre O.



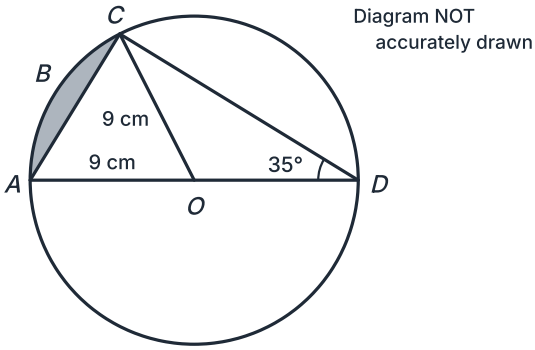
Ikan then adds to this as shown the diagram below.



AF is a diameter.
Find the value of angle OBF, reasons are not needed.

Hint: which angles in the diagram can you find first?

PP 3 **Sydney Boys High School, 2021, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part E (a)** 3 marks



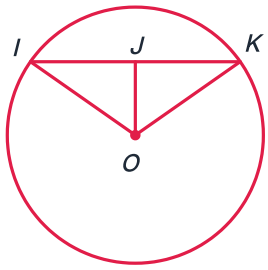
AOD is a diameter of a circle, with centre O and radius 9 cm.
ABC is an arc of the circle.
AC is a chord.
Angle $ADC = 35°$

Calculate the area of the shaded segment.
Give your answer correct to 3 significant figures.

Hint: which angles in the diagram can you find first?

PP 4 **Merewether High School, 2023, Year 10 Assessment Task 4, Mathematics, Section III Part B Q2** 4 marks

O is the centre of the circle and $OJ \perp IK$.
Prove that OJ bisects $\angle IOK$.



Hint: what do you know about the lengths and angles in the diagram?

LESSON C2

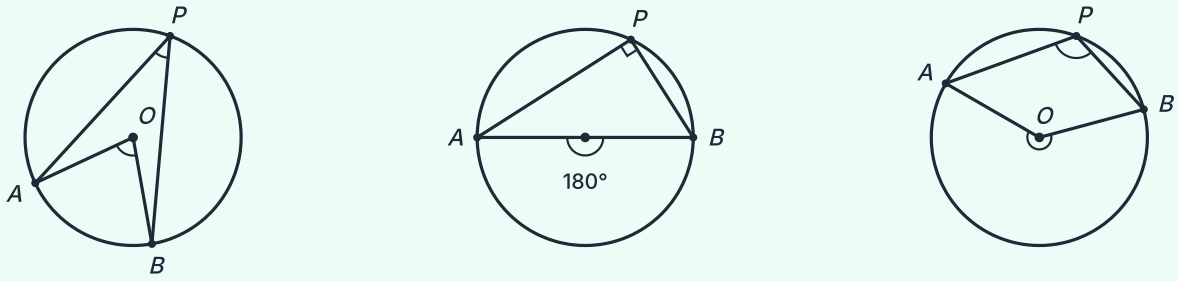
Angles at the centre and circumference

The angle in a semicircle is a special case of a much more general theorem: an angle at the circumference is half the angle at the centre standing on the same arc. From it come two key results: angles at the circumference on the same arc are equal, and the opposite angles of a cyclic quadrilateral are supplementary, with its exterior angle equal to the opposite interior angle. The converse gives a test for a quadrilateral to be cyclic. Two extensions show how circles prove that the altitudes of a triangle are concurrent and what the common ratio in the sine rule really is. The teaching follows the AMSI TIMES module *Circle Geometry*, pages 11 to 17.

CONCEPT 1 · the angle at the centre is twice the angle at the circumference

ANGLES AT THE CENTRE AND CIRCUMFERENCE (AMSI, PAGE 11)

The angle-in-a-semicircle theorem can be generalised considerably. In each diagram below, AB is an arc of a circle with centre O, and P is a point on the opposite arc. The arc AB subtends the angle $\angle AOB$ at the centre. The arc also subtends the angle $\angle APB$, called an **angle at the circumference subtended by the arc AB**.



In the middle diagram, where the arc is a semicircle, the angle at the centre is a straight angle, and by the previous theorem, the angle at the circumference is a right angle – exactly half. We shall show that this relationship holds also for the other two cases, when the arc is a minor arc (left-hand diagram) or a major arc (right-hand diagram). The proof uses isosceles triangles in a similar way to the proof of Thales' theorem.

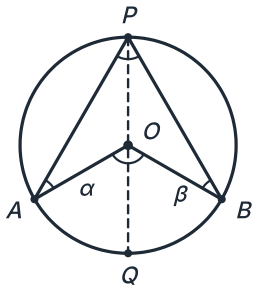
THEOREM

An angle at the circumference of a circle is half the angle at the centre subtended by the same arc.

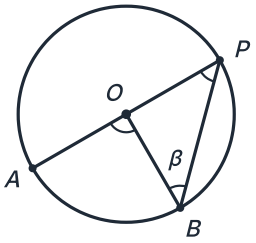
WORKED PROOF **From the text (AMSI, pages 11 and 12): keep for homework**

PROOF

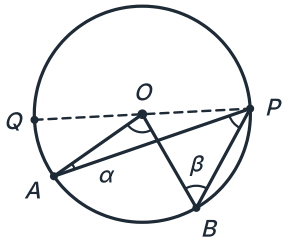
Let AB be an arc of a circle with centre O, and let P be any point on the opposite arc. We need to prove $\angle AOB = 2\angle APB$. The proof divides into three cases, depending on whether:



Case 1: O lies inside $\triangle ABP$



Case 2: O lies on $\triangle APB$



Case 3: O lies outside $\triangle APB$.

Case 1: Join PO and produce PO to Q. Then $OA = OB = OP$ (radii), so we have two isosceles triangles OAP and OBP.

Let $\angle OAP = \alpha$ and $\angle OBP = \beta$.

Then $\angle APO = \alpha$ (base angles of isosceles $\triangle OAP$)

and $\angle BPO = \beta$ (base angles of isosceles $\triangle OBP$).

Hence $\angle AOQ = 2\alpha$ (exterior angle of $\triangle OAP$)

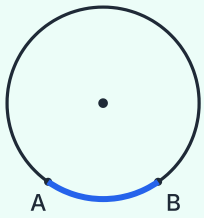
and $\angle BOQ = 2\beta$ so (exterior angle of $\triangle OBP$),

$$\angle AOB = 2\alpha + 2\beta = 2(\alpha + \beta) = 2 \times \angle APB.$$

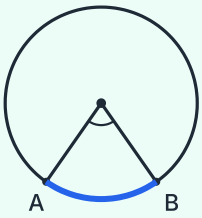
STEPS THE SOURCE SKIPS: CASE 1

The radius PO splits $\angle APB$ into $\angle APO + \angle BPO = \alpha + \beta$, and splits $\angle AOB$ into $\angle AOQ + \angle BOQ$. The exterior angle of $\triangle OAP$ at O equals the sum of the two opposite interior angles, $\alpha + \alpha = 2\alpha$. Cases 2 and 3 are AMSI Exercise 6, Medium homework Question 12.

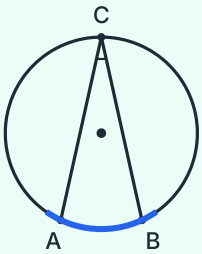
ANGLES SUBTENDED BY AN ARC (MATHS PATHWAY)



Arc AB

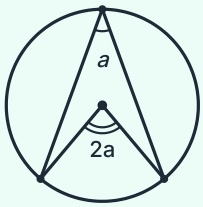


Angle **subtended** by arc AB at the centre



Angle **subtended** by arc AB at point C

Angles subtended at the centre. The angle subtended by an arc at the centre of a circle is twice the angle subtended by the same arc at a point on the circle.



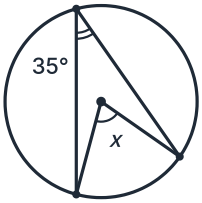
REFLEX ANGLES AT THE CENTRE

If the angle at the circumference is obtuse, it stands on a major arc, and the angle at the centre on that arc is a **reflex** angle, more than $180°$. The theorem still holds: an angle of $110°$ at the circumference goes with a reflex angle of $220°$ at the centre. Angles in a revolution at the centre add to $360°$.

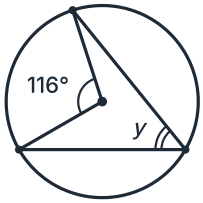
TUTOR DEMO **The angle at the centre**

Calculate the values of the variables in the following circle figures. O marks the centre.

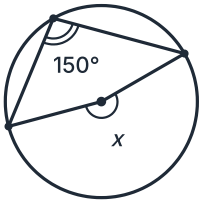
a



b



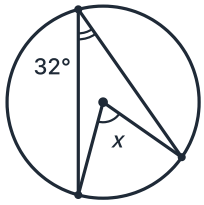
c



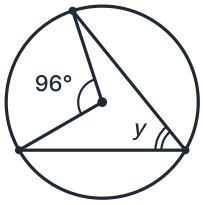
YOUR TURN **Same method**

Calculate the values of the variables in the following circle figures. O marks the centre.

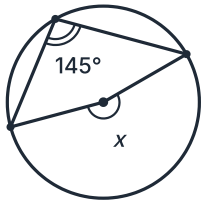
a



b

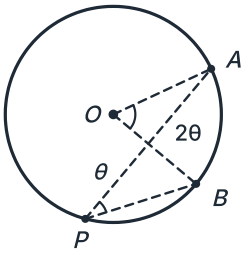


c



WORKED EXAMPLE **From the text (AMSI, page 12): keep for homework**

A punter stands on the edge of a circular racing track. With his binoculars he is following a horse that is galloping around the track at one revolution a minute. Explain why the punter's binoculars are rotating at a constant rate of half a revolution per minute.

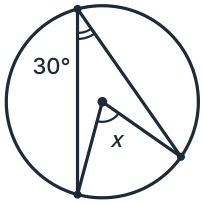


SOLUTION

As the horse moves from position A to position B, the horse moves an angle $\angle AOB = 2\theta$ about the centre O of the track. The angle at the circumference is half the angle at the centre, so $\angle APB = \theta$ which means that the punter's binoculars rotate by an angle θ the horse moves from A to B. Hence the punter is rotating his binoculars at a constant rate that is half the rate at which the horse is rotating about the centre.

WORKED EXAMPLE **Maths Pathway, Arcs and Angles in Circles, Q3(a): keep for homework**

Calculate the values of the variables in the following circle figures.



SOLUTION

$$x = 2 \times 30^\circ$$
$$= 60^\circ$$

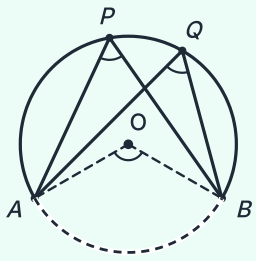
STEPS THE SOURCE SKIPS: Q3(A)

The $30°$ angle at the circumference and the angle x at the centre stand on the same arc (the lower arc between the two lower points), so x is twice $30°$.

CONCEPT 2 · angles on the same arc

ANGLES SUBTENDED BY THE SAME ARC (AMSI, PAGE 12)

In the diagram to the right, the two angles $\angle APB$ and $\angle AQB$ are subtended by the same (minor) arc AB. Each angle is half the angle $\angle AOB$ at the centre subtended by the same arc, so $\angle APB = \angle AQB$.



This corollary of the previous theorem is a particularly significant result about angles in circles:

THEOREM

Two angles at the circumference subtended by the same arc are equal.

Thales' theorem is a special case of this theorem.

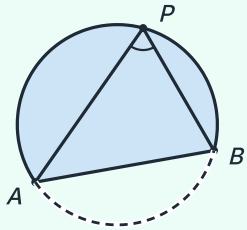
SOME ALTERNATIVE TERMINOLOGY (AMSI, PAGE 13)

The last two theorems are often expressed in slightly different language, and some explanation is needed to avoid confusion.

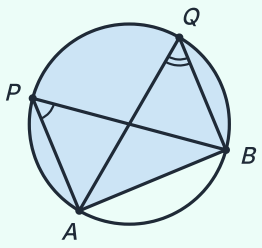
1. An angle subtended by an arc is often said to be **standing on the arc**. With this terminology, the two theorems become:
   - An angle at the circumference of a circle is half the angle at the centre standing on the same arc.
   - Two angles at the circumference standing on the same arc are equal.

   In the context of these two theorems, it is best to avoid the phrases 'standing on a chord AB' and 'subtended by a chord AB', because we need to distinguish between angles subtended by the major arc AB and angles subtended by the minor arc AB.
2. In the upper diagram to the right, the angle $\angle APB$ is called an angle in the (major) segment. Notice that it actually stands on the minor arc AB, which can be confusing. We have already used this terminology before in speaking about an 'angle in a semicircle'. With this terminology, the last theorem becomes:

   Two angles in the same segment of a circle are equal.

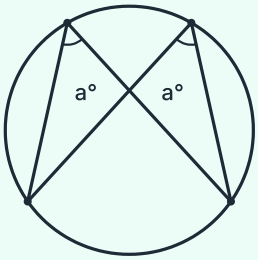


The situation is illustrated in the lower diagram to the right.



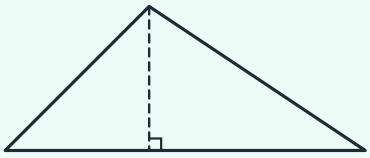
ANGLES SUBTENDED BY THE SAME ARC (MATHS PATHWAY)

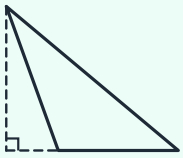
The angles subtended by the same arc at any two points on a circle are equal.



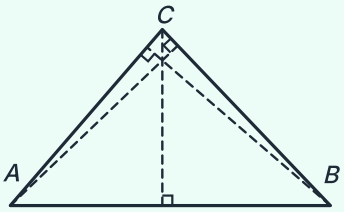
EXTENSION – THE ORTHOCENTRE OF A TRIANGLE (AMSI, PAGE 13)

An **altitude** of a triangle is a perpendicular from any of the three vertices to the opposite side, produced if necessary. The two cases are illustrated in the diagrams below.





There are three altitudes in a triangle. The following theorem proves that they concurrent at a point called the **orthocentre** H of the triangle. It is surprising that circles can be used to prove the concurrence of the altitudes.



Theorem

The altitudes of a triangle are concurrent.

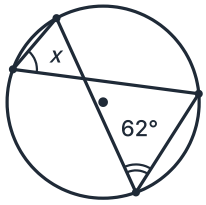
WHERE IS THE PROOF?

Exercise 7 proves this theorem: Hard homework Question 14.

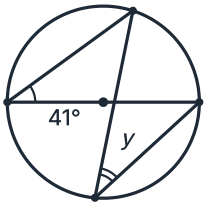
TUTOR DEMO **Angles on the same arc**

Calculate the values of the variables in the following circle figures.

a



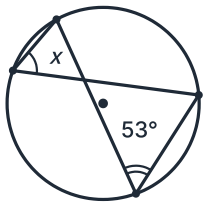
b



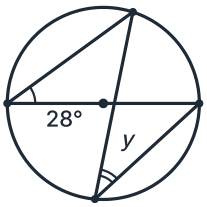
YOUR TURN **Same method**

Calculate the values of the variables in the following circle figures.

a

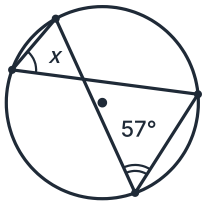


b



WORKED EXAMPLE **Maths Pathway, Arcs and Angles in Circles, Q4(a): keep for homework**

Calculate the values of the variables in the following circle figures.



SOLUTION

$x = 57°$

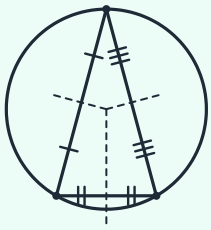
STEPS THE SOURCE SKIPS: Q4(A)

The angle x and the $57°$ angle are both at the circumference, and both stand on the same arc (the upper-right arc between the top point and the right-hand point). So they are equal. The centre plays no part here.

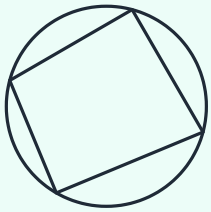
CONCEPT 3 · cyclic quadrilaterals

CYCLIC QUADRILATERALS (AMSI, PAGE 14)

In the module, *Congruence*, we showed how to draw the **circumcircle** through the vertices of any triangle. To do this, we showed that the perpendicular bisectors of its three sides are concurrent, and that their intersection, called the **circumcentre** of the triangle, is equidistant from each vertex.

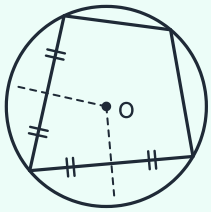


No other circle passes through these three vertices. If we tried to take as centre a point P other than the circumcentre, then P would not lie on one of the perpendicular bisectors, so it could not be equidistant from the three vertices.



When there are four points, we can always draw a circle through any three of them (provided they are not collinear), but only in special cases will that circle pass through the fourth point. A **cyclic quadrilateral** is a quadrilateral whose vertices all lie on a circle. This is the last type of special quadrilateral that we shall consider.

Constructing the circumcircle of a cyclic quadrilateral. Suppose that we are given a quadrilateral that is known to be cyclic, but whose circumcentre is not shown (perhaps it has been rubbed out). The circumcentre of the quadrilateral is the circumcentre of the triangle formed by any three of its vertices, so the construction to the right will find its circumcentre.



The opposite angles of a cyclic quadrilateral. The distinctive property of a cyclic quadrilateral is that its opposite angles are supplementary. The following proof uses the theorem that an angle at the circumference is half the angle at the centre standing on the same arc.

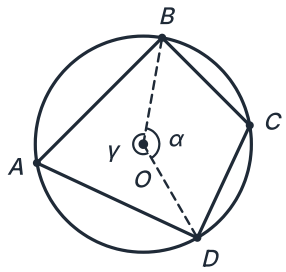
THEOREM

The opposite angles of a cyclic quadrilateral are supplementary.

WORKED PROOF **From the text (AMSI, page 15): keep for homework**

PROOF

Let $ABCD$ be a cyclic quadrilateral with O the centre of the circle.
Join the radii OB and OD. Let α and γ be the angles at the centre, as shown on the diagram.



$$\begin{aligned} \text{Then} \quad & \alpha + \gamma = 360^\circ && \text{(angles in a revolution at } O\text{)} \\ \text{Also} \quad & \angle A = \tfrac{1}{2}\alpha && \text{(angles on the same arc } BCD\text{)} \\ \text{and} \quad & \angle C = \tfrac{1}{2}\gamma && \text{(angles on the same arc } BAD\text{)} \\ \text{so} \quad & \angle A + \angle C = \tfrac{1}{2}\alpha + \tfrac{1}{2}\gamma = 180^\circ \end{aligned}$$

Hence also $\angle ABC + \angle ADC = 180^\circ$ (angle sum of quadrilateral $ABCD$)

Here is an alternative proof using the fact that two angles in the same segment are equal.

STEPS THE SOURCE SKIPS: THE OPPOSITE ANGLES

α is the angle at the centre standing on the arc BCD, and $\angle A$ stands on the same arc, so $\angle A = \frac{1}{2}\alpha$. Then $\frac{1}{2}\alpha + \frac{1}{2}\gamma = \frac{1}{2}(\alpha + \gamma) = \frac{1}{2} \times 360^\circ = 180^\circ$. The angles of any quadrilateral add to 360°, so the other pair also adds to $360^\circ - 180^\circ = 180^\circ$. The alternative proof is AMSI Exercise 8, Medium homework Question 8.

EXTERIOR ANGLES OF A CYCLIC QUADRILATERAL (AMSI, PAGE 15)

An exterior angle of a cyclic quadrilateral is supplementary to the adjacent interior angle, so is equal to the opposite interior angle. This gives us the corollary to the cyclic quadrilateral theorem:

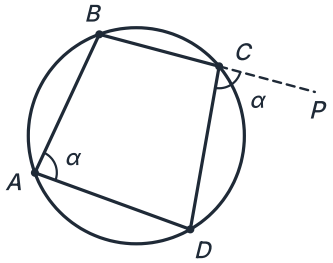
THEOREM

An exterior angle of a cyclic quadrilateral is equal to the opposite interior angle.

WORKED PROOF **From the text (AMSI, page 15): keep for homework**

PROOF

In the diagram to the right, BC is produced to P to form the exterior angle $\angle PCD$. This exterior angle and $\angle A$ are both supplementary to $\angle BCD$, so they are equal.



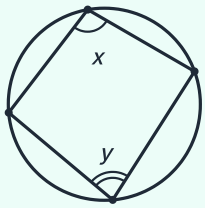
STEPS THE SOURCE SKIPS: THE EXTERIOR ANGLE

$\angle PCD + \angle BCD = 180°$ (angles on the straight line BCP) and $\angle A + \angle BCD = 180°$ (opposite angles of the cyclic quadrilateral). So $\angle PCD = 180° - \angle BCD = \angle A$.

QUADRILATERALS IN CIRCLES (MATHS PATHWAY)

If all four points of a quadrilateral are on a circle, the sum of opposite angles is $180°$.

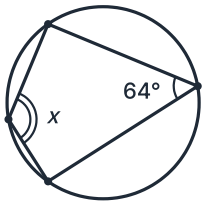
$$x + y = 180°$$



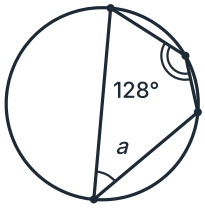
TUTOR DEMO **Cyclic quadrilaterals**

Calculate the value of the variables in the following circle figures.

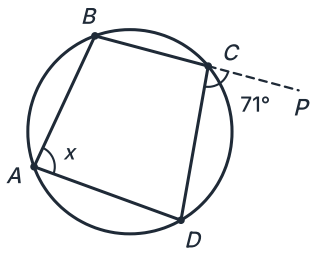
a



b



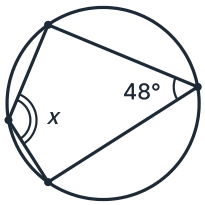
c



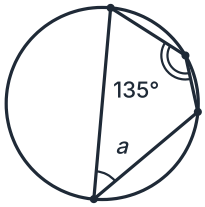
YOUR TURN **Same method**

Calculate the value of the variables in the following circle figures.

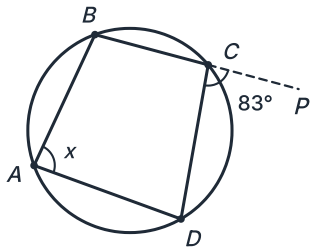
a



b

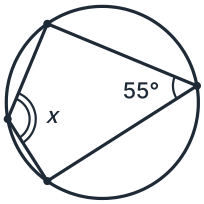


c



WORKED EXAMPLE **Maths Pathway, Arcs and Angles in Circles, Q6(a): keep for homework**

Calculate the value of the variables in the following circle figures.



SOLUTION

$$x = 180° - 55°$$
$$= 125°$$

STEPS THE SOURCE SKIPS: Q6(A)

All four vertices are on the circle, so the quadrilateral is cyclic, and x and the $55°$ angle are opposite angles. Opposite angles of a cyclic quadrilateral add to $180°$.

CONCEPT 4 · a test for a cyclic quadrilateral

EXTENSION – A TEST FOR A CYCLIC QUADRILATERAL (AMSI, PAGE 16)

The property of a cyclic quadrilateral proven earlier, that its opposite angles are supplementary, is also a test for a quadrilateral to be cyclic. That is the converse is true. This theorem completes the structure that we have been following – for each special quadrilateral, we establish its distinctive properties, and then establish tests for it.

The proof uses ‘proof by contradiction’, and is thus a little more difficult than other Year 10 material.

THEOREM

If the opposite angles of a quadrilateral are supplementary, then the quadrilateral is cyclic.

WORKED PROOF **From the text (AMSI, page 16): keep for homework**

PROOF

Let $ABCD$ be a quadrilateral with $\angle A + \angle C = 180°$.

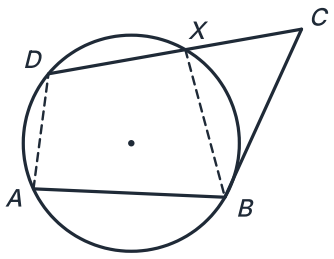
Construct the circle through A, B and D, and suppose, by way of contradiction, that the circle does not pass through C.

Let DC, produced if necessary, meet the circle again at X, and join XB.

Then $\angle A$ and $\angle BXD$ are supplementary because $ABXD$ is a cyclic quadrilateral,

so the angles $\angle DXB$ and $\angle DCB$ are equal, so $XB \parallel CB$

Hence XB and CB are the same line, so C and X coincide, that is the circle does pass through C.



STEPS THE SOURCE SKIPS: THE CONVERSE

Both $\angle DXB$ and $\angle DCB$ are $180° - \angle A$. They are corresponding angles on the line DXC, so $XB \parallel CB$. Two parallel lines through the same point B must be the same line, which forces $X = C$. Medium homework Question 10 (AMSI Exercise 11) proves the same test in exterior-angle form.

TUTOR DEMO **Is the quadrilateral cyclic?**

A quadrilateral $ABCD$ has $\angle A = 97°$, $\angle B = 78°$ and $\angle C = 83°$. Find $\angle D$, and decide whether $ABCD$ is a cyclic quadrilateral, giving a reason.

YOUR TURN **Same method**

A quadrilateral $PQRS$ has $\angle P = 88°$, $\angle Q = 96°$ and $\angle R = 93°$. Find $\angle S$, and decide whether $PQRS$ is a cyclic quadrilateral, giving a reason.

CONCEPT 5 · extension, the sine rule and circle geometry

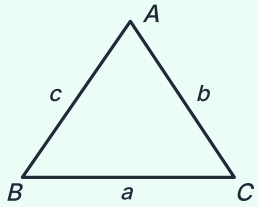
EXTENSION – THE SINE RULE AND CIRCLE GEOMETRY (AMSI, PAGE 17)

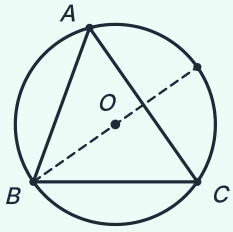
The sine rule states that for any triangle ABC, the ratio of any side over the sine of its opposite angle is a constant,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Each term is the ratio of a length over a pure number, so their common value seems to be a length. Thus it reasonable to ask, what is this common length? The answer is a surprise – the common length is the diameter of the circumcircle through the vertices of the triangle.

The proof of this result provides a proof of the sine rule that is independent of the proof given in the module, *Further Trigonometry*.





THEOREM

In any triangle ABC, $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$, where R is the radius of the circumcircle.

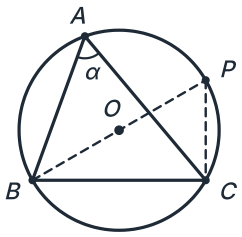
WORKED PROOF **From the text (AMSI, page 17): keep for homework**

PROOF

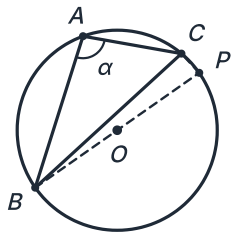
Let O be the centre of the circumcircle through A, B and C, and let $\angle A = \alpha$.

It is sufficient to prove that $\dfrac{a}{\sin A}$ is the diameter of the circumcircle.

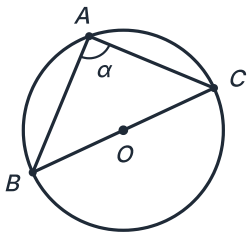
There are three cases, as shown below.



Case 1: A and O lie on the same side of BC.



Case 2: A and O lie on opposite sides of BC.



Case 3: O lies on BC

In cases 1 and 2, construct the diameter BOP, and join PC.

FINISHING THE PROOF

Exercise 13 completes the three cases: Hard homework Question 15.

TUTOR DEMO **The circumradius**

In $\triangle ABC$, $a = 12$ cm and $A = 30°$. Find the diameter and the radius of the circle through A, B and C.

YOUR TURN **Same method**

In $\triangle ABC$, $a = 9$ cm and $A = 50°$. Find the radius of the circle through A, B and C, correct to two decimal places.

HOMEWORK · Maths Pathway sheet, AMSI exercises and past papers

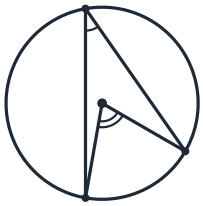
Each question is from the source named above it. Answers are in the answers booklet. In every figure, a dot inside the circle marks its centre.

EASY

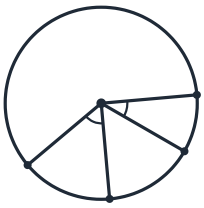
1 Maths Pathway, Arcs and Angles in Circles, Q1

In each of the following figures, are the two marked angles subtended by the same arc?

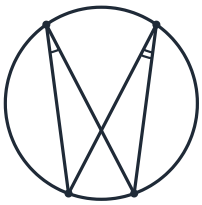
a



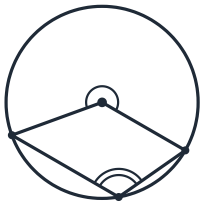
b



c



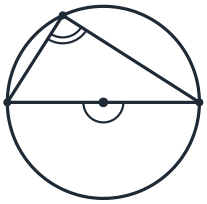
d



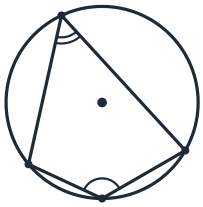
e



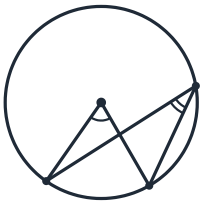
f



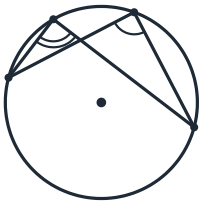
g



h



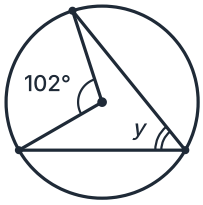
i



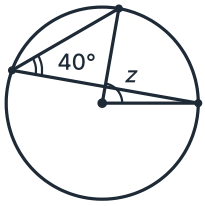
2 Maths Pathway, Arcs and Angles in Circles, Q3(b)–(f)

Calculate the values of the variables in the following circle figures.

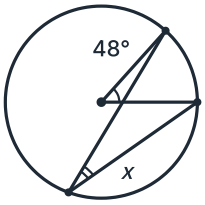
b



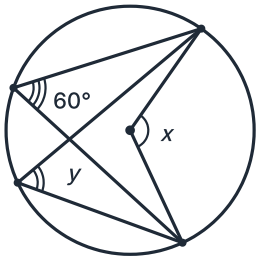
c



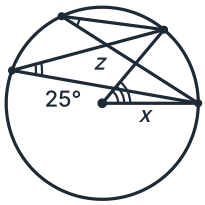
d



e



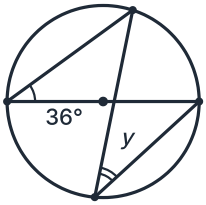
f



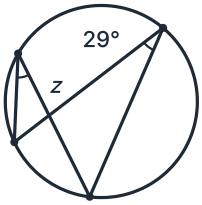
3 Maths Pathway, Arcs and Angles in Circles, Q4(b)–(c)

Calculate the values of the variables in the following circle figures.

b



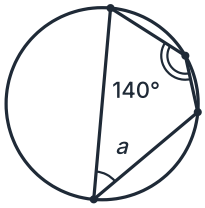
c



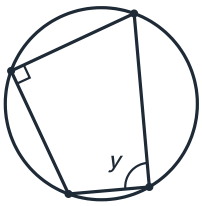
4 Maths Pathway, Arcs and Angles in Circles, Q6(b)–(c)

Calculate the value of the variables in the following circle figures.

b



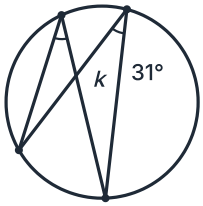
c



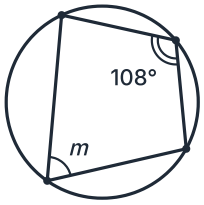
5 Maths Pathway, Arcs and Angles in Circles, Q8(b)–(d)

Calculate the values of the variables in the following circle figures, and explain your answer.

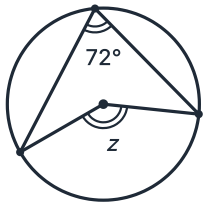
b



c



d

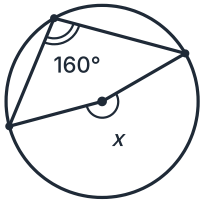


MEDIUM

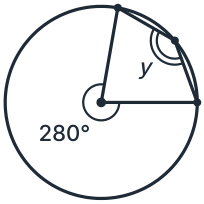
6 Maths Pathway, Arcs and Angles in Circles, Q5

Calculate the values of the variables in the following circle figures.

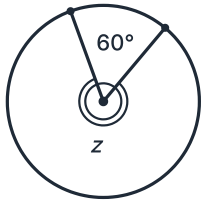
a



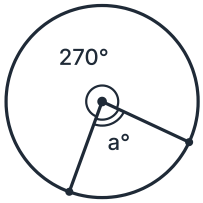
b



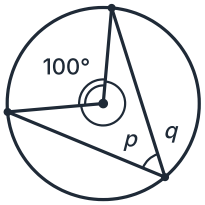
c



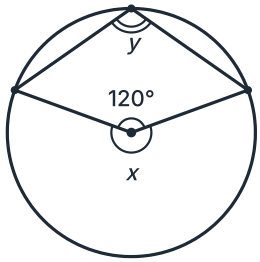
d



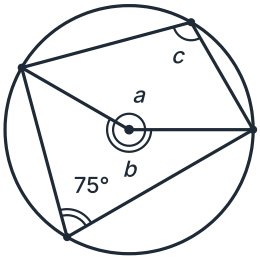
e



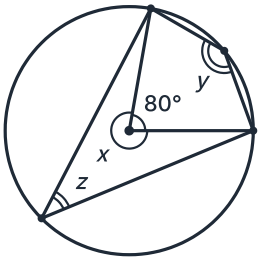
f



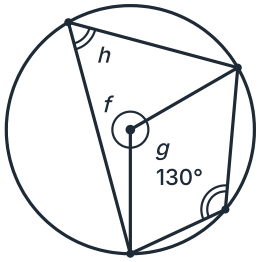
g



h

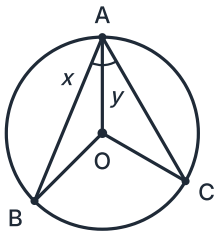


i



7 Maths Pathway, Arcs and Angles in Circles, Q2

In this question, you will prove that $\angle BOC = 2 \times \angle BAC$.



a Show that ΔOAB and ΔOAC in the figure to the right are isosceles.

b Use what you proved in part (a) to write $\angle OBA$ and $\angle OCA$ in terms of x and y.

c Use what you found in part (b) to write $\angle AOB$ and $\angle AOC$ in terms of x and y.

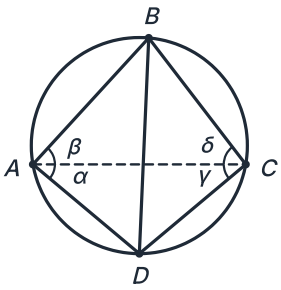
d Write $\angle BOC$ in terms of x and y.

e Write $\angle BAC$ in terms of x and y.

f Use what you found in parts (d) and (e) to write $\angle BOC$ in terms of $\angle BAC$.

8 AMSI TIMES Circle Geometry, Exercise 8

Join the diagonals AC and BD of the cyclic quadrilateral $ABCD$. Let α, β, γ and δ be the angles shown.



a Give a reason why $\angle DBC = \alpha$

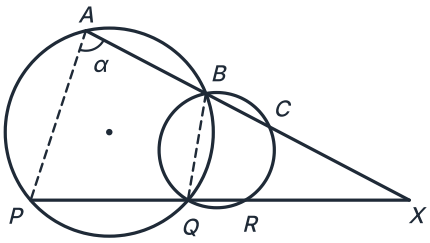
b What other angles have sizes β, γ and δ?

c Prove that $\alpha + \beta + \gamma + \delta = 180°$.

d Hence prove that the opposite angles of $ABCD$ are supplementary.

9 AMSI TIMES Circle Geometry, Exercise 9

Show that $AP \parallel CR$ in the diagram to the right.



10 AMSI TIMES Circle Geometry, Exercise 11

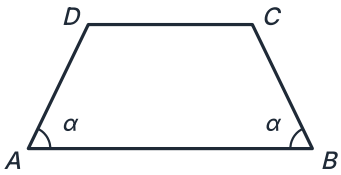
Prove the following alternative form of the above theorem:

If an exterior angle of a quadrilateral equals the opposite interior angle, then the quadrilateral is cyclic.

'The above theorem' is the test for a cyclic quadrilateral in Concept 4: if the opposite angles of a quadrilateral are supplementary, then the quadrilateral is cyclic.

11 AMSI TIMES Circle Geometry, Exercise 12

In the diagram to the right, the two adjacent acute angles of the trapezium are equal.
Prove that the trapezium is cyclic.



12 AMSI TIMES Circle Geometry, Exercise 6

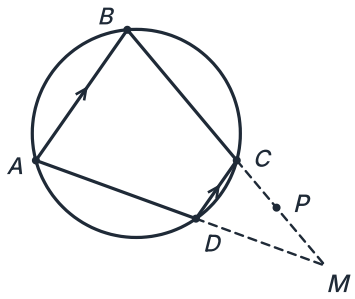
Complete the proof in the other two cases.

The cases are those of the proof in Concept 1: Case 2, O on $\triangle APB$, and Case 3, O outside $\triangle APB$. Their diagrams are with that proof.

HARD

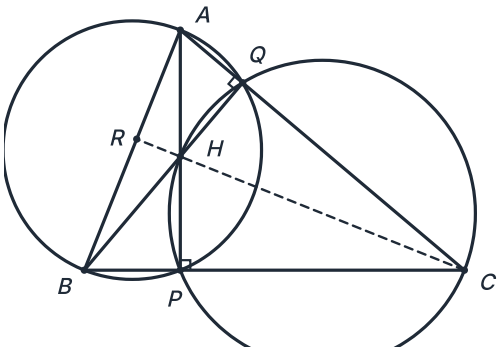
13 AMSI TIMES Circle Geometry, Exercise 10

If a cyclic trapezium is not a rectangle, show that the other two sides are not parallel, but have equal length.



14 AMSI TIMES Circle Geometry, Exercise 7

In the diagram to the right, the altitudes AP and BQ meet at H. The interval CH is produced to meet AB, produced if necessary, at R. We need to prove that $CR \perp AB$.



a Explain why Q, H, P and C are concyclic, and draw the circle.

b Explain why A, B, P and Q are concyclic, and draw the circle.

c Join the common chord PQ, and explain why $\angle QCH = \angle QPH$.

d Explain why $\angle APQ = \angle ABQ$.

e Use $\triangle ABQ$ and $\triangle ACR$ to explain why $CR \perp AB$.

15 AMSI TIMES Circle Geometry, Exercise 13

a Complete the following steps of the proof in Case 1.

i Explain why $\angle BCP = 90°$ and $\angle P = \alpha$

ii Complete the proof using $\triangle BCP$.

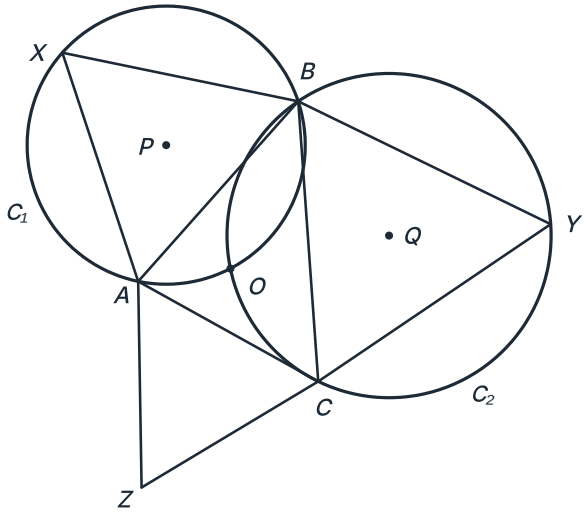
b Complete the following steps of the proof in Case 2.

i Explain why $\angle BCP = 90°$ and $\sin P = \sin \alpha$

ii Complete the proof using $\triangle BCP$.

c Complete the proof in Case 3.

16 **Sydney Boys High School, 2021, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part G (e)** **5 marks**



A scalene triangle ABC has equilateral triangles ABX, BCY and ACZ constructed on its three sides. Circle $\mathcal{C}_1$ is drawn to pass through the vertices of $\triangle ABX$ and has centre P. Similarly circle $\mathcal{C}_2$ is drawn through the vertices of $\triangle BCY$ and has centre Q. These two circles meet at B and O as shown in the diagram above.

i Find the sizes of $\angle AOB$, $\angle BOC$, and $\angle AOC$. 1

ii Prove that $AOCZ$ is a cyclic quadrilateral. 1

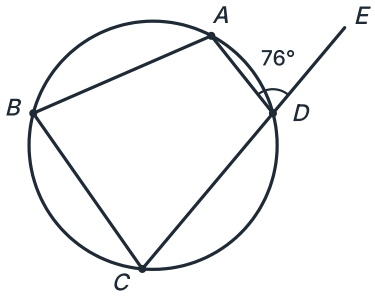
iii Taking R as the centre of circle $AOCZ$, prove that ΔPQR is equilateral. 3

Hint: which angles in the diagram do you know already?

PAST-PAPER HOMEWORK · mini exam

Suggested time: 25 minutes **17 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Sydney Boys High School, 2021, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part B (p)** **1 mark**

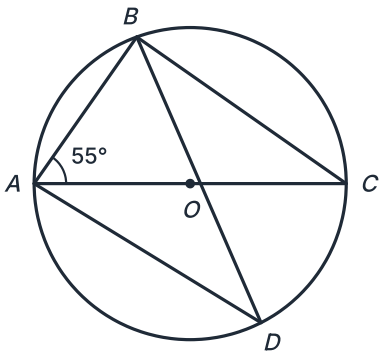


A, B, C, and D are concyclic points.
CDE is a straight line.
Write down the value of $\angle ABC$.

Hint: what do you know about the points A, B, C and D?

PP 2 **Sydney Boys High School, 2023, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part C Q4** **3 marks**

In the diagram below, A, B, C, and D are points on a circle, centre O.
AOC is a diameter of the circle.
$\angle BAC = 55°$.
Find $\angle ADB$, giving full reasons.

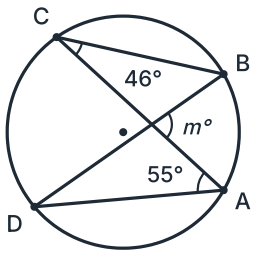


NOT TO SCALE

Hint: which angles in the diagram can you find first?

PP 3 **Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q25(b)** 4 marks

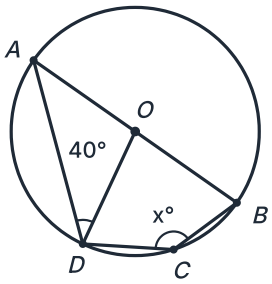
Find the value of the pronumeral giving reasons.



Hint: which angles in the diagram can you find first?

PP 4 **Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q25(c)** 4 marks

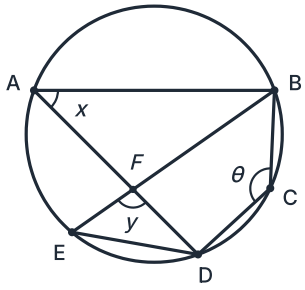
O is the centre of the circle. Calculate the value of x, giving reasons.



Hint: what do you know about the lengths and angles in the diagram?

PP 5 **Sydney Boys High School, 2024, Year 10 Stage 5 Term 4 Yearly Exam, Mathematics, Part E Q2** 5 marks

In the diagram below, $ABCDE$ is a circle. It is given that $AB \parallel ED$.
AD and BE intersect at the point F.



i Show that $2x + y = 180$, giving reasons. 3

ii Express x in terms of θ. 1

iii Express y in terms of θ. 1

Hint: which given facts about the diagram can you use first?

LESSON C3

Tangents and secants

A tangent meets a circle at just one point, and it is perpendicular to the radius there. From this come the equal tangents from an external point, and the alternate segment theorem, which links the angle between a tangent and a chord to the angles in the other segment. The lesson ends with three theorems that combine similarity with circles: intersecting chords, secants from an external point, and the square on a tangent. The teaching follows the AMSI TIMES module *Circle Geometry*, pages 18 to 27. A chapter review across Lessons C1 to C3 follows the past-paper set.

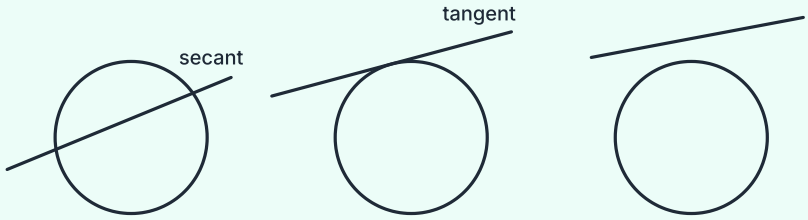
CONCEPT 1 · the tangent and the radius

TANGENTS TO CIRCLES (AMSI, PAGE 18)

A **tangent** to a circle is a line that meets the circle at just one point. The diagram below shows that given a line and a circle, three possibilities can arise:

- The line may be a **secant**, cutting the circle at two points.
- The line may be a **tangent**, touching the circle at just one point.
- The line may miss the circle entirely.

The words ‘secant’ and ‘tangent’ are from Latin – ‘secant’ means ‘cutting’ (as in ‘cross-section’), and ‘tangent’ means ‘touching’ (as in ‘tango’).



The point where a tangent touches a circle is called a **point of contact**. It is not immediately obvious how to draw a tangent at a particular point on a circle, or even whether there may be more than one tangent at that point. The following theorem makes the situation clear, and uses Pythagoras’ theorem in its proof.

THEOREM

Let T be a point on a circle with centre O.

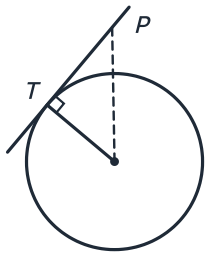
a The line through T, perpendicular to the radius OT, is a tangent to the circle.
b This line is the only tangent to the circle at T.
c Every point on the tangent, except for T itself, lies outside the circle.

WORKED PROOF **From the text (AMSI, page 19): keep for homework**

PROOF

First we prove parts **a** and **c**. Let ℓ be the line through T perpendicular to the radius OT. Let P be any other point on ℓ, and join the interval OP. Then using Pythagoras' theorem in $\triangle OTP$,

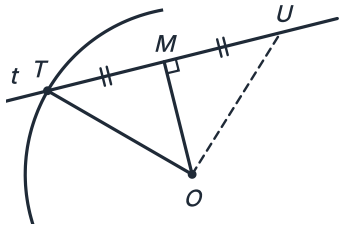
$OP^2 = OT^2 + PT^2$, which is greater than OT^2,



so OP is greater than the radius OT. Hence P lies outside the circle, and not on it. This proves that the line ℓ is a tangent, because it meets the circle only at T. It also proves that every point on ℓ, except for T, lies outside the circle.

It remains to prove part **b**, that there is no other tangent to the circle at T.

Let t be a tangent at T, and suppose, by way of contradiction, that t were not perpendicular to OT. Let M be the foot of the perpendicular from O to t, and construct the point U on t, on the other side of M, so that $UM = MT$. Then using Pythagoras' theorem in $\triangle OMT$ and $\triangle OMU$,



$OT^2 = OM^2 + MT^2 = OM^2 + MU^2 = OU^2$,

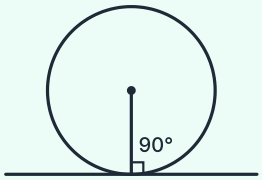
So $OU = OT$. Hence U also lies on the circle, contradicting the fact that t is a tangent.

STEPS THE SOURCE SKIPS: THE TANGENT THEOREM

$PT > 0$ because P is a different point from T, so $OT^2 + PT^2$ is greater than OT^2. In part **b**, $MT = MU$ by the construction, which is why $OM^2 + MT^2 = OM^2 + MU^2$. A tangent meets the circle only once, so a second point U on the circle is impossible.

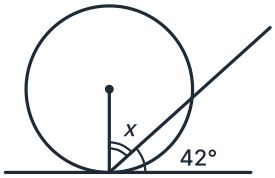
ANGLE BETWEEN TANGENT AND RADIUS (MATHS PATHWAY)

The tangent to a circle is always perpendicular to the radius.

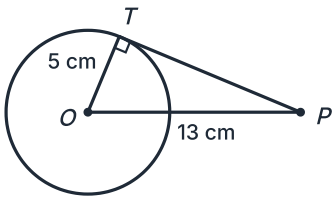


TUTOR DEMO **Tangent and radius**

a Find the value of the missing variable, giving reasons.

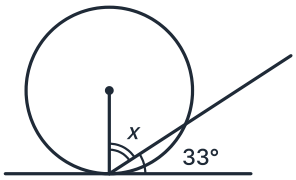


b PT is a tangent to the circle with centre O and radius 5 cm, and $OP = 13$ cm. Find the length of PT.

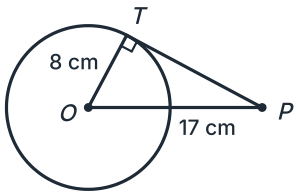


YOUR TURN **Same method**

a Find the value of the missing variable, giving reasons.

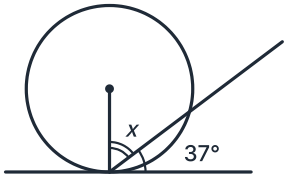


b PT is a tangent to the circle with centre O and radius 8 cm, and $OP = 17$ cm. Find the length of PT.



WORKED EXAMPLE **Maths Pathway, Chords, Secants and Tangents, Q8(a): keep for homework**

Find the value of the missing variable, giving reasons.



SOLUTION

The tangent is perpendicular to the radius, therefore

$x + 37° = 90°$

$x = 53°$

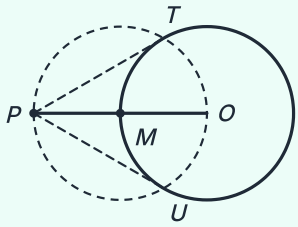
STEPS THE SOURCE SKIPS: Q8(A)

The radius and the tangent make a right angle at the point of contact. The line from that point splits the right angle into x and $37°$, so $x = 90° - 37°$.

CONCEPT 2 · tangents from an external point

CONSTRUCTION – TANGENTS FROM AN EXTERNAL POINT (AMSI, PAGE 19)

Using this radius and tangent theorem, and the angle in a semicircle theorem, we can now construct tangents to a circle with centre O from a point P outside the circle.



1. Join OP and construct the midpoint M of OP.
2. Draw the circle with centre M through O and P, and let it meet the circle at T and U.

The angles $\angle PTO$ and $\angle PUO$ are right angles, because they are angles in a semicircle. Hence the lines PT and PU are tangents, because they are perpendicular to the radii OT and OU, respectively.

Tangents from an external point have equal length. It is also a simple consequence of the radius-and-tangent theorem that the two tangents PT and PU have equal length.

Notice that here, and elsewhere, we are using the word ‘tangent’ in a second sense, to mean not the whole line, but just the interval from an external point to the point of contact.

THEOREM

Tangents to a circle from an external point have equal length.

WHERE IS THE PROOF?

AMSI sets it as Exercise 14 (‘Prove this result using either congruence or Pythagoras’ theorem’): Medium homework Question 9.

TANGENTS AND TRIGONOMETRY (AMSI, PAGE 20)

The right angle formed by a radius and tangent gives further opportunities for simple trigonometry. The following exercise shows how the names ‘tangent’ and ‘secant’, and their abbreviations $\tan\theta$ and $\sec\theta$, came to be used in trigonometry. In a circle of radius 1, the length of a tangent subtending an angle θ at the centre is $\tan\theta$, and the length of the secant from the external point to the centre is $\sec\theta$.

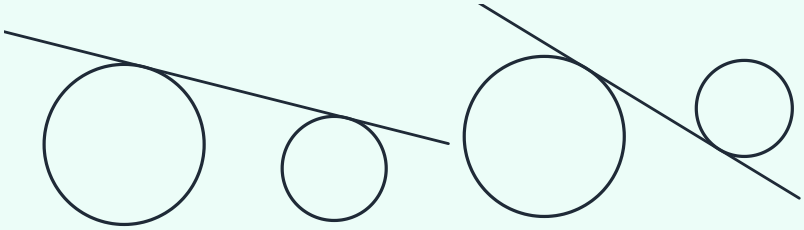
A NAME, NOT A NEW FUNCTION

The function $\sec\theta = \dfrac{1}{\cos\theta}$ is not part of the Year 10 course; it is named here only to explain where the word 'tangent' in trigonometry comes from. In right-angled $\triangle OTP$ with $OT = 1$, $\tan\theta = \dfrac{PT}{OT} = PT$. The exercise the box refers to (AMSI Exercise 15) uses $\sec\theta$, so it is not in this pack.

QUADRILATERALS WITH INCIRCLES, COMMON TANGENTS AND TOUCHING CIRCLES (AMSI, PAGES 20 AND 21)

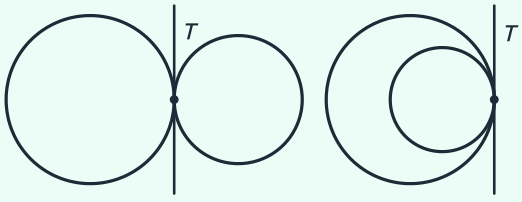
The following exercise involves quadrilaterals within which an incircle can be drawn tangent to all four sides. These quadrilaterals form yet another class of special quadrilaterals.

A line that is tangent to two circles is called a **common tangent** to the circles. When the points of contact are distinct, there are two cases, as in the diagrams below. The two circles lie on the same side of a **direct common tangent**, and lie on opposite sides of an **indirect common tangent**.



Direct common tangent Indirect common tangent

Two circles are said to touch at a common point T if there is a common tangent to both circles at the point T. As in the diagram below, the circles **touch externally** when they are on opposite sides of the common tangent, and **touch internally** when they are on the same side of the common tangent.



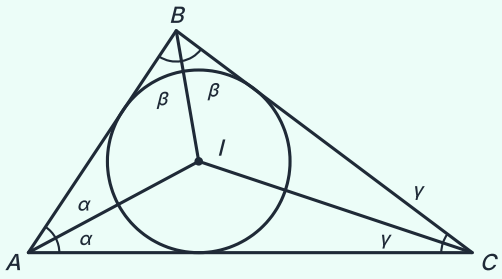
Provided that they are distinct, touching circles have only the one point in common.

WHERE ARE THESE EXERCISES?

The exercise on quadrilaterals with incircles is AMSI Exercise 16, Medium homework Question 10. Common tangents and touching circles are AMSI Exercise 17, homework Questions 11 (Medium) and 15 (Hard).

EXTENSION – THE INCENTRE OF A TRIANGLE (AMSI, PAGES 21 AND 22)

We are now in a position to prove a wonderful theorem on the angle bisectors of a triangle. These three bisectors are concurrent, and their point of intersection is called the **incentre** of the triangle. The incentre is the centre of the **incircle** tangent to all three sides of the triangle, as in the diagram to the right.



Theorem. The angle bisectors of a triangle are concurrent, and the resulting incentre is the centre of the incircle, that is tangent to all three sides.

Proof. We have to prove IC bisects $\angle ACB$.
Given $\triangle ABC$, construct the angle bisectors of $\angle A$ and $\angle B$, and let I be their point of intersection.
Join IC, and let $\alpha = \angle BAI = \angle CAI$ and $\beta = \angle ABI = \angle CBI$.
Drop perpendiculars IL, IM and IN to BC, CA and AB respectively.

Summary – The four centres of a triangle. This completes the development of the four best-known centres of a triangle. The results, and the associated terminology and notation, are summarised here for reference.

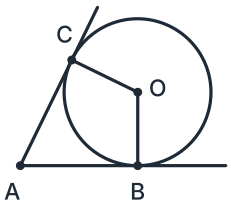
- The perpendicular bisectors of the three sides of a triangle are concurrent at the **circumcentre** O, which is the centre of the **circumcircle** through all three vertices. This was proven in the module, *Congruence*.
- The angle bisectors of the three angles of a triangle are concurrent at the **incentre** I, which is the centre of the **incircle** that is tangent to all three sides. This was proven above.
- A median of a triangle is an interval joining a vertex to the midpoint of an opposite side. The medians of a triangle are concurrent at the **centroid** G, which divides each median in the ratio $2:1$. This was proven in the module, *Scale Drawings and Similarity*.
- An altitude of a triangle is a perpendicular from a vertex to the opposite side, produced if necessary. The altitudes of a triangle, also produced if necessary, are concurrent at the **orthocentre** H. This was proven earlier in the present module as an extension in the section on angles at the centre and circumference.

FINISHING THE INCENTRE PROOF

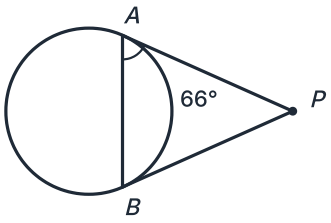
AMSI completes the proof in Exercise 18, Medium homework Question 12.

TUTOR DEMO **Two tangents from a point**

a AB and AC are tangents to the circle with centre O, and $\angle CAB = 64°$. Find $\angle COB$.

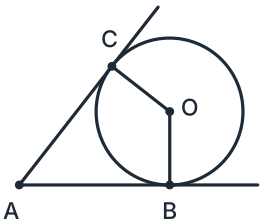


b PA and PB are tangents, and the angle between the chord AB and the tangent PA is $66°$. Find $\angle APB$.

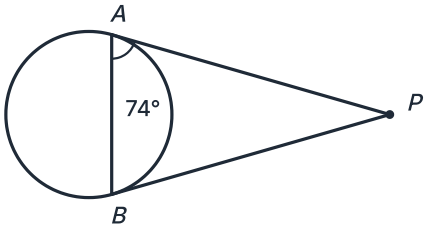


YOUR TURN **Same method**

a AB and AC are tangents to the circle with centre O, and $\angle CAB = 52°$. Find $\angle COB$.

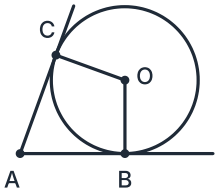


b PA and PB are tangents, and the angle between the chord AB and the tangent PA is $74°$. Find $\angle APB$.



WORKED EXAMPLE **Maths Pathway, Chords, Secants and Tangents, Q9(a): keep for homework**

Copy the figure. Then, use the information provided to find the required value. Show your working.
Given information: $\angle CAB = 70°$
Find the value of $\angle COB$.



SOLUTION

Since AB is tangent to the circle, $\angle OBA = 90°$
Since AC is tangent to the circle, $\angle OCA = 90°$
The sum of angles in a quadrilateral is $360°$, therefore

$$\angle COB + \angle CAB + \angle OBA + \angle OCA = 360°$$
$$\angle COB + 70° + 90° + 90° = 360°$$
$$\angle COB = 110°$$

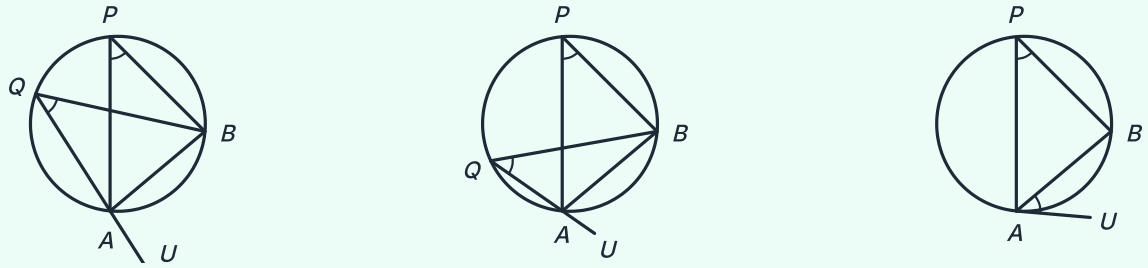
STEPS THE SOURCE SKIPS: Q9(A)

$70° + 90° + 90° = 250°$, and $360° - 250° = 110°$. The two right angles come from the radii OB and OC meeting the tangents at the points of contact.

CONCEPT 3 · the alternate segment theorem

THE ALTERNATE SEGMENT THEOREM (AMSI, PAGE 22)

The left-hand diagram below shows two angles $\angle P$ and $\angle Q$ lying in the same segment of a circle – we have proven that these two angles are equal. In the next two diagrams, the angle $\angle BQU$ remains equal to $\angle P$ as the point Q moves around the arc closer and closer to A. In the last diagram, Q coincides with A, and AU is a tangent.



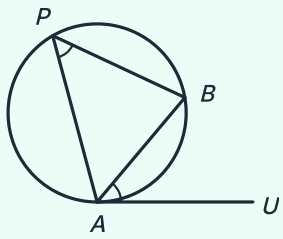
It seems reasonable from these diagrams that in the limiting case, the angle $\angle BAU$ will still equal $\angle P$ – this is what the alternate segment theorem says.

THEOREM

An angle between a chord and a tangent is equal to any angle in the alternate segment.

THE ALTERNATE SEGMENT (AMSI, PAGE 23)

Here $\angle BAU$ is 'an angle between a chord and a tangent'. The word 'alternate' means 'other' – the chord AB divides the circle into two segments, and the **alternate segment** is the segment on the left containing the angle $\angle P$. We have already proven that all the angles in this left-hand segment are equal.

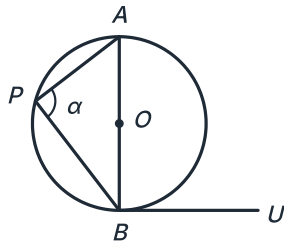


There are two equally satisfactory proofs of this theorem. One is written out below and the other is left as an exercise.

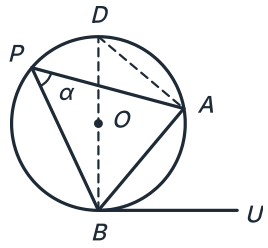
WORKED PROOF **From the text (AMSI, pages 23 and 24): keep for homework**

PROOF

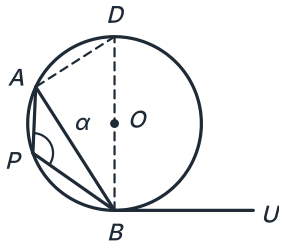
Let AB be a chord of a circle and let BU be a tangent at B.
Let P be a point on the arc that is not within the arms of $\angle ABU$, and let $\angle P = \alpha$.
There are three cases, depending on whether:
Case 1: The centre O lies on the chord AB.
Case 2: The centre O lies outside the arms of $\angle ABU$.
Case 3: The centre O lies within the arms of $\angle ABU$.
In each case, we need to prove that $\angle ABU = \alpha$



Case 1



Case 2



Case 3

Case 1 is easily settled:

$\alpha = 90°$ (angle in a semicircle) and $\angle ABU = 90°$ (radius and tangent)

For the other two cases, construct the diameter BOD, and join DA.

Case 2:	$\angle ADB = \alpha$	(angles on the same arc AB)
and	$\angle BAD = 90°$	(angle in a semicircle)
so	$\angle DBA = 90° - \alpha$	(angle sum of $\triangle ABD$)
But	$\angle DBU = 90°$	(radius and tangent)
so	$\angle ABU = \alpha$	(adjacent angles at B)
Case 3:	$\angle ADB = 180° - \alpha$	(cyclic quadrilateral $APBD$)
and	$\angle BAD = 90°$	(angle in a semicircle)
so	$\angle ABD = \alpha - 90°$	(angle sum of $\triangle ABD$)

But	$\angle DBU = 90°$	(radius and tangent)
so	$\angle ABU = \alpha$	(adjacent angles at B)

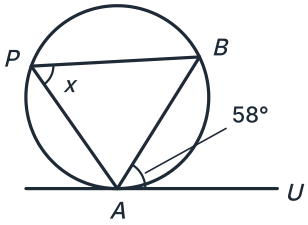
STEPS THE SOURCE SKIPS: THE ALTERNATE SEGMENT THEOREM

In Case 2, $\angle ABU = \angle DBU - \angle DBA = 90° - (90° - \alpha) = \alpha$. In Case 3, the ray BD lies outside the angle $\angle ABU$, so $\angle ABU = \angle ABD + \angle DBU = (\alpha - 90°) + 90° = \alpha$. The other proof, with the radii AO and BO, is AMSI Exercise 19, Hard homework Question 16.

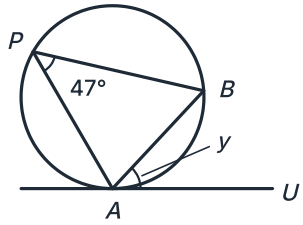
TUTOR DEMO **The alternate segment**

AU is a tangent to the circle at A. Find the value of each pronumeral, giving reasons.

a



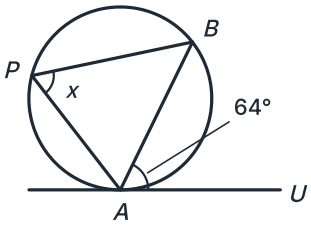
b



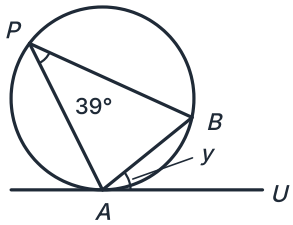
YOUR TURN **Same method**

AU is a tangent to the circle at A. Find the value of each pronumeral, giving reasons.

a



b



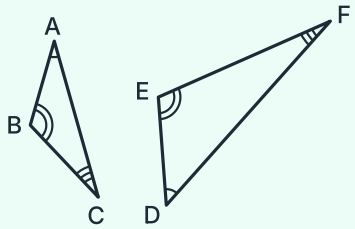
CONCEPT 4 · intersecting chords

RECALL: SIMILAR TRIANGLES (MATHS PATHWAY)

Similar triangles have the following two properties:

- Corresponding angles are equal.
- The ratios of corresponding sides are the same.

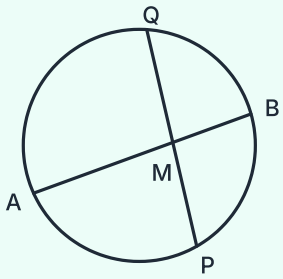
You can use either property to show two triangles are similar.



$\triangle ABC$ is similar to $\triangle DEF$: $\angle BAC = \angle EDF$, $\angle ABC = \angle DEF$, $\angle ACB = \angle DFE$, and $\dfrac{AB}{DE} = \dfrac{BC}{EF} = \dfrac{AC}{DF}$

SIMILARITY AND CIRCLES: INTERSECTING CHORDS (AMSI, PAGE 24)

The final theorems in this module combine similarity with circle geometry to produce three theorems about intersecting chords, intersecting secants, and the square on a tangent.



The first theorem deals with chords that intersect within the circle. In this situation, each chord cuts the other into two sub-intervals called **intercepts**. It is an amazing consequence of similar triangles that, in this situation, the products of the intercepts on each chord are equal. That is, in the diagram to the right, $AM \times MB = PM \times MQ$.

THEOREM

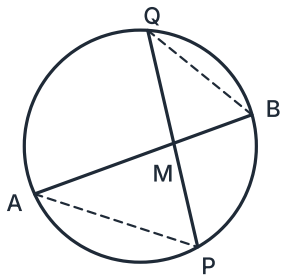
When two chords intersect within a circle, the products of the intercepts are equal.

WORKED PROOF **From the text (AMSI, page 25): keep for homework**

PROOF

Let AB and PQ be chords intersecting at M. Join AP and BQ.

In the triangles APM and QBM:



$\angle PAM = \angle BQM$ (angles on the same arc PB)

$\angle APM = \angle QBM$ (angles on the same arc AQ)

so $\triangle APM$ is similar to $\triangle QBM$ (AA):

Hence $\dfrac{AM}{QM} = \dfrac{PM}{BM}$ (matching sides of similar triangles)

so $AM \times BM = PM \times QM$.

The very last step is particularly interesting. It converts the equality of two ratios of lengths to the equality of two products of lengths. This is a common procedure when working with similarity.

STEPS THE SOURCE SKIPS: INTERSECTING CHORDS

Multiply both sides of $\dfrac{AM}{QM} = \dfrac{PM}{BM}$ by $QM \times BM$: the left side becomes $AM \times BM$ and the right side becomes $PM \times QM$.

THE SINE RULE AND SIMILARITY (AMSI, PAGE 25)

Many problems involving similarity can be handled using the sine rule. The exercise below gives an alternative proof of the intersecting chord theorem using the sine rule to deal directly with the ratio of two sides of the triangles.

The remaining two theorems of this section also have alternative proofs using the sine rule.

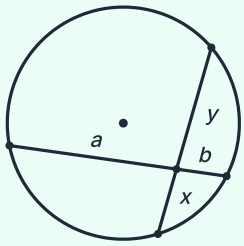
WHERE IS THE EXERCISE?

It is AMSI Exercise 21, Medium homework Question 14.

INTERSECTING CHORDS (MATHS PATHWAY)

Since intersecting chords form similar triangles, you can apply the following rule:

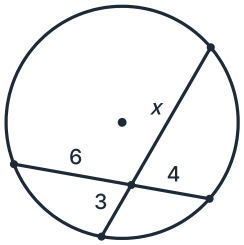
$$ab = xy$$



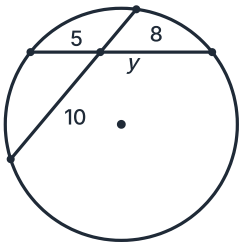
TUTOR DEMO **Intersecting chords**

Find the value of the missing variable in the following figures involving intersecting chords.

a



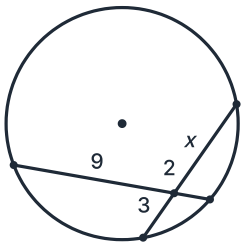
b



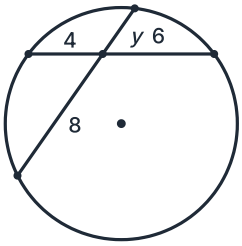
YOUR TURN **Same method**

Find the value of the missing variable in the following figures involving intersecting chords.

a



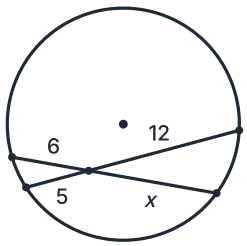
b



WORKED EXAMPLE

Maths Pathway, Intersecting Chords and Secants, Q3(a): keep for homework

Find the value of the missing variable in the following figures involving intersecting chords.



SOLUTION

$$x \times 6 = 5 \times 12$$
$$x = \frac{60}{6}$$
$$= 10$$

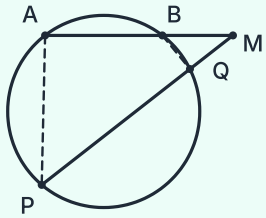
STEPS THE SOURCE SKIPS: Q3(A)

Pair the two intercepts that lie on the same chord: 6 and x are on one chord, 5 and 12 on the other. Then divide both sides by 6.

CONCEPT 5 · secants and tangents from an external point

SECANTS FROM AN EXTERNAL POINT (AMSI, PAGE 25)

Now suppose that the chords do not meet within the circle. If they meet on the circle, the identity above holds trivially, and if they are parallel, there is nothing to say. Otherwise, we can produce the chords until they intersect outside the circle, and an analogous theorem applies.



We are now dealing with secants from an external point. When a secant through an external point M meets a circle at two points A and B, the lengths AM and BM are called the **intercepts** of the secant from the external point, and as before $AM \times MB = PM \times MQ$.

Be careful here, the chord AB is not one of the two intercepts. With this definition of intercept, the previous theorem can now be stated as follows. Its proof is word for word almost the same, apart from the reasons:

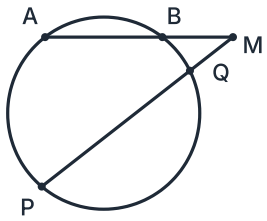
THEOREM

The product of the intercepts on two secants from an external point are equal.

WORKED PROOF **From the text (AMSI, page 26): keep for homework**

PROOF

Let ABM and PQM be two secants from an external point M, as shown.
Join AP and BQ.



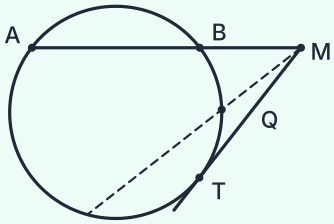
FINISHING THE PROOF

AMSI leaves the rest as an exercise. Medium homework Question 13 (AMSI Exercise 23) proves it with AQ and BP joined instead, and Question 8 proves it with other letters.

TANGENT AND SECANT FROM AN EXTERNAL POINT (AMSI, PAGE 26)

Now imagine the secant MPQ in the previous diagrams rotating until it becomes a tangent at a point T on the circle. As the secant rotates, the length of each intercept PM and QM gets closer to the length of the tangent TM to the circle from M, so the product $PM \times QM$ gets closer to the square TM^2 of the tangent from M. This is a proof using limits.

When we draw a secant and a tangent from M, we have seen that the product $AM \times BM$ equals the square TM^2 of the tangent. The geometric proof is similar to the previous two proofs, but it does require the alternate segment theorem to establish the similarity.



THEOREM

The product of the intercepts on a secant from an external point equals the square of the tangent from that point.

WORKED PROOF **From the text (AMSI, page 27): keep for homework**

PROOF

Let ABM be a secant, and TM a tangent, from an external point M, as shown. We need to prove that $AM \times BM = TM^2$. Join AT and BT.

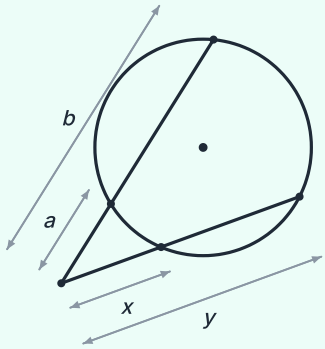
FINISHING THE PROOF

AMSI completes it in Exercise 24, Hard homework Question 18, which has the diagram.

INTERSECTING SECANTS (MATHS PATHWAY)

Since intersecting secants form similar triangles, you can apply the following rule:

$$ab = xy$$

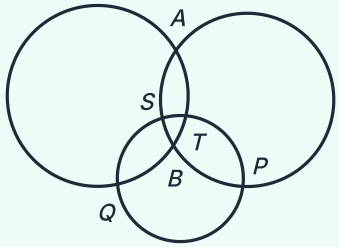


OUTER PART TIMES WHOLE SECANT

For a secant from an external point, the two intercepts are the outer part (from the point to the near intersection) and the whole secant (from the point to the far intersection). For the tangent and secant theorem, the tangent counts twice: $TM \times TM = AM \times BM$.

THE CONCURRENCE OF THREE COMMON CHORDS (AMSI, PAGE 27)

Proving concurrence usually involves logic that is a little more sophisticated than required in most proofs. The following theorem is a difficult application of the intersecting chord theorem.



LINKS FORWARD, AND THE CONVERSE OF THE CIRCLE THEOREMS (AMSI, PAGE 27)

Circle geometry is often used as part of the solution to problems in trigonometry and calculus.

The circle theorems proven in this module all have dramatic and important converse theorems, which are tests for points to lie on a circle. The proofs of these converses, and their applications, are usually regarded as inappropriate for Years 9–10, apart from the converse of the angle in a semicircle theorem, which was developed within the module. They are so closely related to the material in this module, however, that they have been fully developed in an Appendix.

THE CONCURRENCE EXERCISE

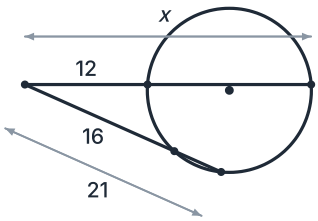
'The following theorem' is AMSI Exercise 25, Hard homework Question 19.

CONVERSES IN THIS PACK

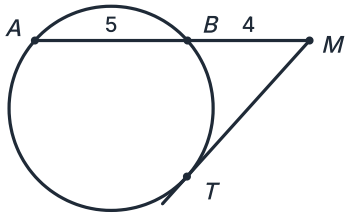
The AMSI appendix is not part of this pack. You have met two converses: the converse of the angle in a semicircle (Lesson C1) and the test for a cyclic quadrilateral (Lesson C2).

TUTOR DEMO **Secants and tangents from a point**

a Calculate the value of the missing variable.

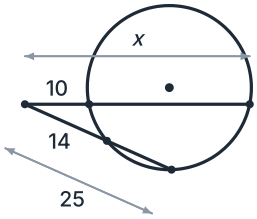


b MT is a tangent at T, $MB = 4$ and $BA = 5$. Find the length of MT.

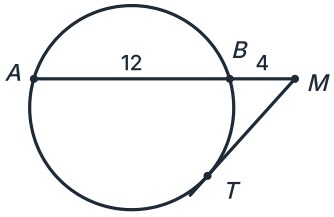


YOUR TURN **Same method**

a Calculate the value of the missing variable.



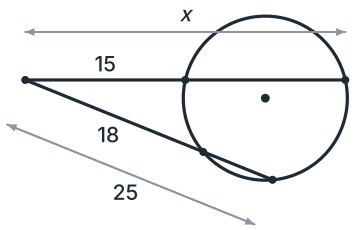
b MT is a tangent at T, $MB = 4$ and $BA = 12$. Find the length of MT.



WORKED EXAMPLE

Maths Pathway, Intersecting Chords and Secants, Q6(a): keep for homework

Calculate the value of the missing variable.



SOLUTION

$$\begin{aligned} 15 \times x &= 18 \times 25 \\ x &= \frac{18 \times 25}{15} \\ &= 6 \times 5 \\ &= 30 \end{aligned}$$

STEPS THE SOURCE SKIPS: Q6(A)

On the upper secant the outer part is 15 and the whole secant is x; on the lower secant the outer part is 18 and the whole secant is 25. $\frac{18}{15} = \frac{6}{5}$, so $\frac{18 \times 25}{15} = \frac{6 \times 25}{5} = 6 \times 5$.

HOMEWORK · Maths Pathway sheets, AMSI exercises and a past paper

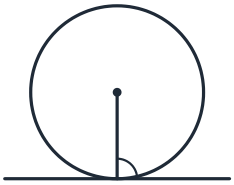
Each question is from the source named above it. Answers are in the answers booklet. In every figure, a dot inside the circle marks its centre.

EASY

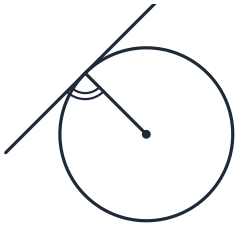
1 Maths Pathway, Chords, Secants and Tangents, Q7

Use a protractor to measure the angle between the tangent and the radius in each of the following figures:

a

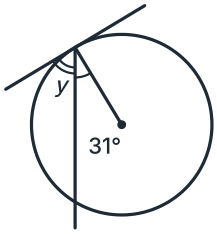


b



2 Maths Pathway, Chords, Secants and Tangents, Q8(b)

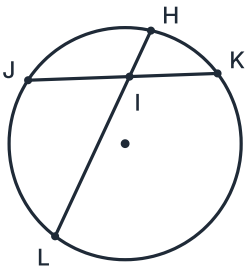
Find the value of the missing variable, giving reasons.



3 Maths Pathway, Intersecting Chords and Secants, Q2

Use the ratio property of similar triangles to find the missing values.

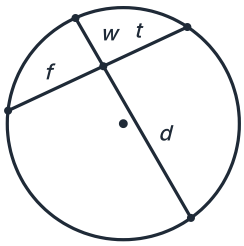
a



Since ΔIHJ is similar to ΔIKL,

$$\frac{HI}{\square} = \frac{JI}{\square}$$ and $HI \times \square = JI \times \square$

b

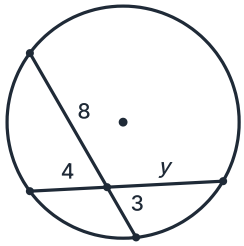


$f \times \square = \square \times \square$

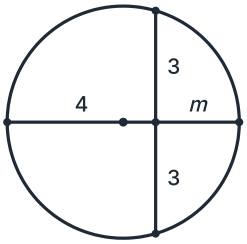
4 Maths Pathway, Intersecting Chords and Secants, Q3(b)–(d)

Find the value of the missing variable in the following figures involving intersecting chords.

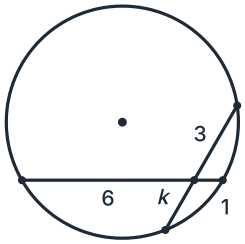
b



c



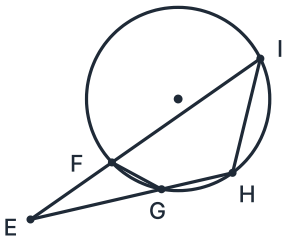
d



5 Maths Pathway, Intersecting Chords and Secants, Q5

Use the ratio property of similar triangles to find the missing values.

a

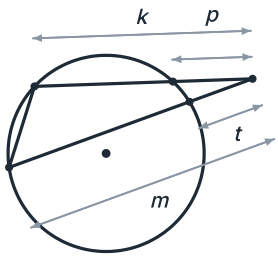


Since ΔEFG is similar to ΔEHI,

$\frac{EF}{\square} = \frac{EG}{\square}$ and

$EF \times \square = EG \times \square$

b

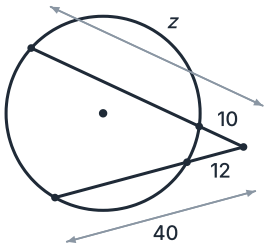


$p \times \square = \square \times \square$

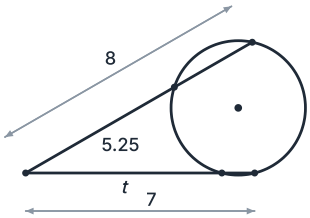
6 Maths Pathway, Intersecting Chords and Secants, Q6(b)–(d)

Calculate the value of the missing variable.

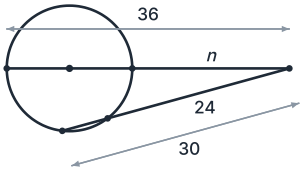
b



c



d



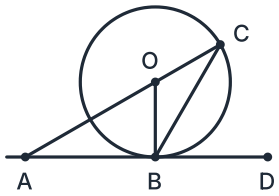
MEDIUM

7 Maths Pathway, Chords, Secants and Tangents, Q9(b)

Copy the figure. Then, use the information provided to find the required value. Show your working.

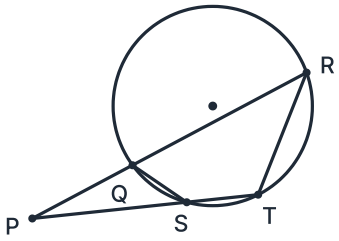
Given information: $\angle CBD = 60°$

Find the value of $\angle OAB$.



8 Maths Pathway, Intersecting Chords and Secants, Q4

This question will use similar triangles to prove a property about secants which intersect at a point outside the circle.



a Prove that ΔPQS is similar to ΔPTR.

Hint: Use the fact that in a quadrilateral whose points are on a circle, the sum of opposite angles is $180°$.

b Use the ratio property of similar triangles to prove that $PQ \times PR = PS \times PT$.

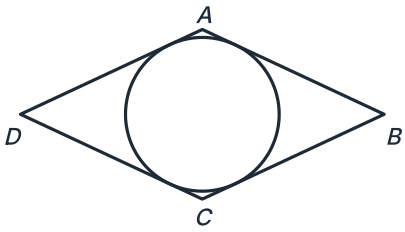
9 AMSI TIMES Circle Geometry, Exercise 14

Prove this result using either congruence or Pythagoras' theorem.

The result is the theorem in Concept 2: tangents to a circle from an external point have equal length.

10 AMSI TIMES Circle Geometry, Exercise 16

The sides of a quadrilateral are tangent to a circle drawn inside it. Show that the sums of opposite sides of the quadrilateral are equal.



11 AMSI TIMES Circle Geometry, Exercise 17(a)

How many common tangents are there to two circles when the circles:

i are one inside the other, **ii** touch internally, **iii** overlap,

iv touch externally, **v** are next to each other without touching?

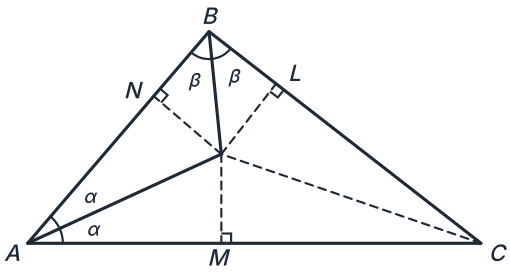
12 AMSI TIMES Circle Geometry, Exercise 18

a Prove that $\triangle MAI \equiv \triangle NAI$ and that $\triangle NBI \equiv \triangle LBI$.

b Explain why $LI = MI = NI$.

c Prove that $\triangle LCI \equiv \triangle MCI$.

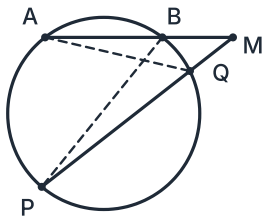
d Hence complete the proof that the angle bisectors meet at I, and that I is the incentre.



13 AMSI TIMES Circle Geometry, Exercise 23

If instead we had joined the intervals AQ and BP, what corresponding changes should be made to the proof?

'The proof' is the secant theorem proof begun in Concept 5, where AP and BQ are joined.



14 AMSI TIMES Circle Geometry, Exercise 21

Use the sine rule in the diagram in the above proof to prove that $\dfrac{AM}{PM} = \dfrac{QM}{BM}$.

The diagram is the one with the proof of the intersecting chord theorem in Concept 4.

HARD

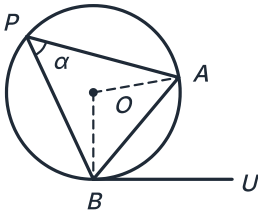
15 AMSI TIMES Circle Geometry, Exercise 17(b)–(c)

b Prove that the intervals between the points of contact on two indirect common tangents have equal length.

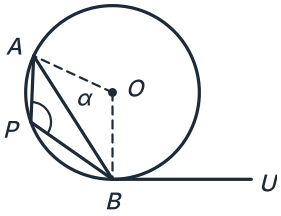
c Prove that the intervals between the points of contact on two direct common tangents have equal length.

16 AMSI TIMES Circle Geometry, Exercise 19

Find an alternative proof in cases 2 and 3 by constructing the radii AO and BO and using angles at the centre.



Case 2

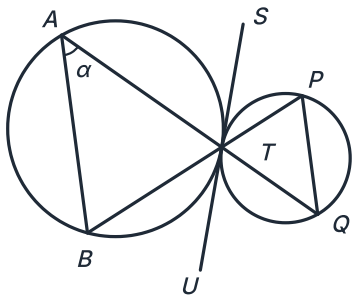


Case 3

17 AMSI TIMES Circle Geometry, Exercise 20

The result in the following exercise is surprising. One would not expect parallel lines to emerge so easily in a diagram with two touching circles.

Prove that $AB \parallel PQ$ in the diagram to the right, where STU is a common tangent to both circles. AQ and BP are straight lines.

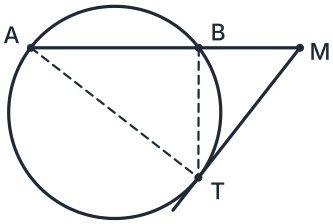


18 AMSI TIMES Circle Geometry, Exercise 24

a Explain why $\angle A = \angle MTB$.

b Prove that $\triangle ATM$ is similar to $\triangle TBM$.

c Complete the proof.



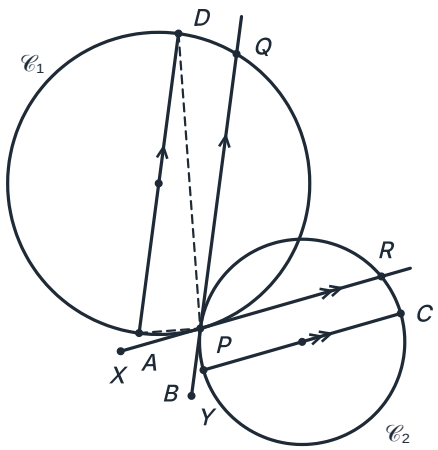
19 AMSI TIMES Circle Geometry, Exercise 25

Show that the three common chords AB, PQ and ST to the three circles in the diagram above are concurrent.

The diagram is the one with 'The concurrence of three common chords' in Concept 5.

20 **Sydney Boys High School, 2023, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part F Q3(a)–(b)** **5 marks**

The diagram shows two circles $\mathcal{C}_1$ and $\mathcal{C}_2$. The point P is one of their points of intersection. The tangent to $\mathcal{C}_2$ at P meets $\mathcal{C}_1$ at Q, and the tangent to $\mathcal{C}_1$ at P meets $\mathcal{C}_2$ at R.



The points A and D are chosen on $\mathcal{C}_1$ so that AD is a diameter of $\mathcal{C}_1$ and parallel to PQ. Likewise, points B and C are chosen on $\mathcal{C}_2$ so that BC is a diameter of $\mathcal{C}_2$ and parallel to PR.

The points X and Y lie on the tangents PR and PQ, respectively, as shown in the diagram.

a Show that $\angle APX = \angle DPQ$. 2

b Show that A, P, and C are collinear. 3

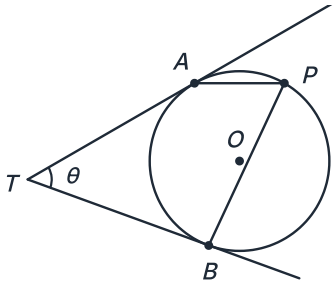
Hint: what do you know about the lines and angles in the diagram?

PAST-PAPER HOMEWORK · mini exam

Suggested time: 25 minutes **18 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Baulkham Hills High School, 2020, Year 10 Yearly Examination, Mathematics, Q9** **1 mark**

The points A, B and P lie on a circle centred at O. The tangents to the circle at A and B meet at the point T, and $\angle ATB = \theta$.



What is $\angle APB$ in terms of θ?

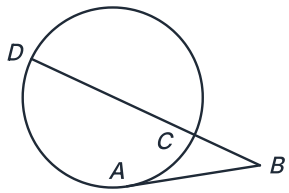
A. $\dfrac{\theta}{2}$ B. $90° - \dfrac{\theta}{2}$

C. θ D. $180° - \theta$

Hint: what do you know about the lines and angles in the diagram?

PP 2 **Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q16** **1 mark**

AB is tangent. $BC = 2$ cm, $CD = 5$ cm. The length of AB in centimetres is



(A) $\sqrt{10}$ (B) $\sqrt{14}$

(C) $\sqrt{20}$ (D) $\sqrt{35}$

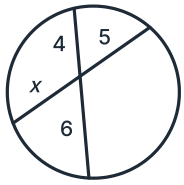
Hint: what do you know about the lengths in the diagram?

PP 3 **Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q17** 1 mark

The value of x is

(A) $3\frac{1}{3}$ (B) $4\frac{4}{5}$

(C) 7 (D) $7\frac{1}{2}$

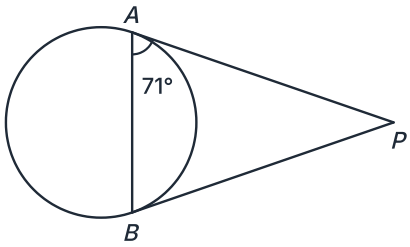


NOT TO SCALE

Hint: what do you know about the lengths in the diagram?

PP 4 **Sydney Boys High School, 2022, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part B Q3** 1 mark

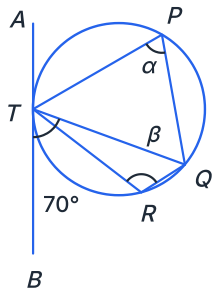
A and B are points on a circle. PA and PB are tangents.
Write down the value of angle APB.



NOT DRAWN TO SCALE

Hint: what do you know about the lengths and angles in the diagram?

PP 5 **Sydney Boys High School, 2020, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics Stage 5.3, Q21** 4 marks



Consider the diagram.

Find the values of the following pronumerals, justifying your answers geometrically.

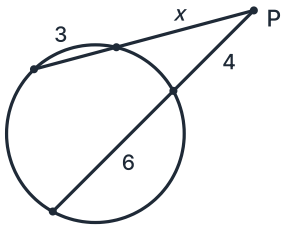
a α 2

b β 2

Hint: which angles in the diagram can you find first?

PP 6 **Baulkham Hills High School, 2020, Year 10 Yearly Examination, Mathematics, Q20** 3 marks

Two secants from the point P intersect a circle as shown in the diagram.



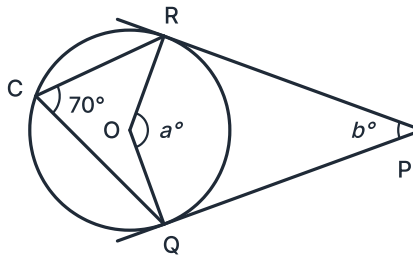
NOT TO SCALE

What is the value of x? Give reasons for your answer.

Hint: what do you know about the lengths in the diagram?

PP 7 **Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q25(a)** 4 marks

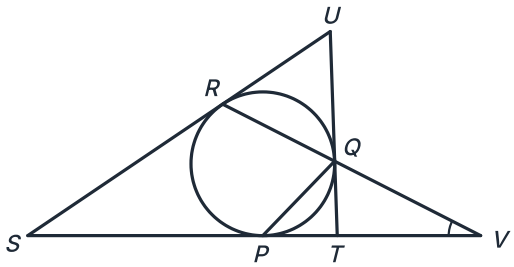
Find the value of the pronumerals giving reasons. O is the centre of the circle.



Hint: which angles in the diagram can you find first?

PP 8 **Sydney Boys High School, 2024, Year 10 Stage 5 Term 4 Yearly Exam, Mathematics, Part C Q7** 3 marks

In the figure below, P, Q and R are points lying on a circle.
Lines ST, TU and SU are the tangents to the circle at P, Q, and R respectively.
RQ produced and ST produced meet at the point V.
If $\angle PSR = 34°$ and $\angle QPT = 46°$, then what is the value of $\angle PVQ$?
Geometrical reasons are needed.



Hint: which angles in the diagram can you find first?

CHAPTER REVIEW

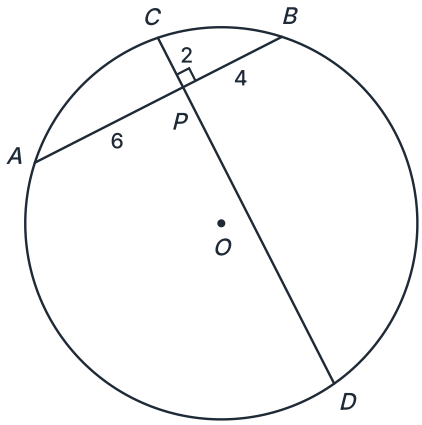
Circle geometry review

Four past-paper questions that mix ideas from the revision and Lessons C1 and C3: chords and the perpendicular from the centre, the angle in a semicircle, isosceles triangles, tangents, the alternate segment theorem and the similarity theorems. Allow about 20 minutes for the 14 marks.

R 1 **Sydney Boys High School, 2020, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics Stage 5.3, Q34** 4 marks

In the diagram below, AB and CD are perpendicular chords that intersect at P on a circle centred at O.

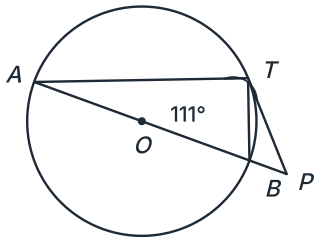
Given that $AP = 6$, $BP = 4$ and $CP = 2$, find the radius of the circle in simplest exact form.



Hint: what do you know about the lengths and angles in the diagram?

R 2 **Sydney Boys High School, 2021, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part F (c)** 2 marks

In the diagram AB is a diameter of the circle, TP is a tangent at point T, O is the centre of the circle and $\angle ATP = 111°$.

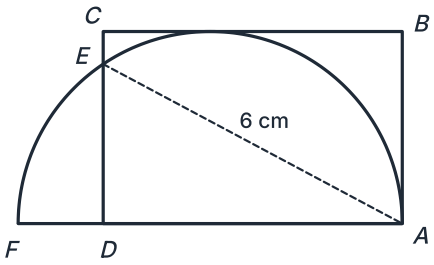


Write down the value of $\angle BAT$, giving clear reasoning.

Hint: which angles in the diagram can you find first?

R 3 **Sydney Boys High School, 2022, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part F Q3** 4 marks

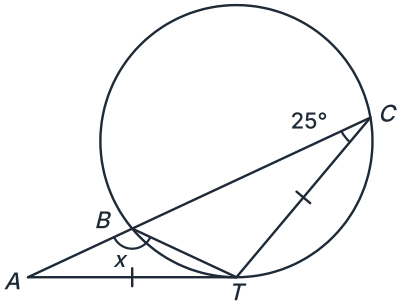
In the diagram below, $ABCD$ is a rectangle and AF is a diameter of a semi-circle with tangent BC. If $AE = 6$ cm, find the area of $ABCD$.



Hint: what do you know about the lines and angles in the diagram?

R 4 **Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q25(d)** 4 marks

AT is a tangent to the circle at T. $TA = TC$. Find the value of x giving reasons.



Hint: what do you know about the lengths and angles in the diagram?